NAIMIRIS: A PROFIT-DRIVEN SMART FARMING MODEL FOR SCOTCH BONNET PEPPERS

INDIVIDUAL THESIS

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NAIMIRIS: A PROFIT-DRIVEN SMART FARMING MODEL FOR SCOTCH BONNET PEPPERS

Individual component: Technology-Assisted A-to-Z Farming Guidance: A Stage-Aware IoT and YOLO-Based Decision-Support Module for Nai Miris (Capsicum chinense) Cultivation

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Abstract

Smallholder Scotch Bonnet (*Nai Miris*; *Capsicum chinense* Jacq.) cultivation requires timely, stage-specific decisions that depend on both visible plant development and root-zone conditions, yet growers rarely have access to plant-level technical advice. This thesis presents the design of the Technology-Assisted A-to-Z Farming Guidance component of NAIMIRIS (R26-IT-103): a stage-aware decision-support workflow that integrates ESP32-based monitoring of air temperature, relative humidity, soil moisture, soil temperature and electrical conductivity with checkpoint smartphone images analysed by stage-specific YOLO detectors (YOLOv8n for seedling emergence, a YOLO flowering detector and YOLO11s for unripe and ripening fruit).

Using a design-science methodology, the work specifies a five-stage crop workflow; validity-aware sensing with an explicit calibration basis; versioned stage profiles with five-band classification and persistence-and-recovery alerting adapted from alarm-management practice; a quality-gated image pipeline; temporal alignment of images with sensor aggregates; and a rule-based fusion engine that produces cautious, evidence-attached recommendations with a complete decision trace. A thirty-case traceable verification suite, a plant-disjoint model-evaluation protocol, a statistical analysis plan and a threats-to-validity analysis are provided.

The thesis separates functionality reported in the project prototype from specified design and from outcomes still to be validated; accordingly, no detector accuracy, sensor accuracy or agronomic benefit is claimed. The contribution is an auditable, interpretable and reproducible workflow that links low-cost monitoring with crop imagery while accounting for sensor validity, image occlusion and crop-specific calibration.

Keywords: Nai Miris; Internet of Things; YOLO; sensor–vision fusion; precision agriculture.

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LIST OF ABBREVIATIONS

Abbreviation	Description
ADC	Analogue-to-Digital Converter
AI	Artificial Intelligence
AP / mAP	Average Precision / mean Average Precision
API	Application Programming Interface
ASMP	Agriculture Sector Modernization Project
COCO	Common Objects in Context (benchmark dataset)
DSR	Design Science Research
EC / ECe	Electrical Conductivity / EC of saturated-paste extract
ESP32	Espressif Systems microcontroller family
FC	Field Capacity
FR / NFR	Functional / Non-Functional Requirement
HORDI	Horticultural Crop Research and Development Institute
IoT	Internet of Things
IoU	Intersection over Union
IWMI	International Water Management Institute
JWT	JSON Web Token
MAE / RMSE	Mean Absolute Error / Root Mean Square Error
MoSCoW	Must, Should, Could, Won't (prioritisation)
NMS	Non-Maximum Suppression
OWASP	Open Worldwide Application Security Project
PDPA	Personal Data Protection Act (Sri Lanka)
REST	Representational State Transfer
RH	Relative Humidity
SUS	System Usability Scale
UTC	Coordinated Universal Time
YOLO	You Only Look Once

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1 INTRODUCTION

This chapter introduces the research problem addressed by the individual component of the NAIMIRIS project, positions it within the wider project and the Sri Lankan agricultural context, states the research gap, aim, objectives and questions, and defines the scope, contribution and structure of the thesis. Throughout, the thesis distinguishes carefully between what has been designed, what has been demonstrated in the project prototype, and what remains to be validated experimentally.

1.1 Background and Motivation

Chilli peppers are among the most economically significant horticultural crops in Sri Lanka, and national policy documents identify dry-chilli production as a priority for import substitution and rural income generation [1], [2]. Within this sector, the Scotch Bonnet or Habanero type (locally *Nai Miris*; *Capsicum chinense* Jacq.) is a comparatively high-value crop with export potential, but it is also demanding to grow: it is sensitive to temperature, water supply and nutrition, and its yield and fruit quality depend on timely decisions at each phase of the crop cycle [3], [4].

Most Sri Lankan growers are smallholders. They typically rely on personal experience, neighbours and occasional extension visits, and have limited access to crop-specific technical advice at the moment a decision is needed. National climate-smart agriculture initiatives and digital advisory platforms have begun to address this gap [5]–[7], but generic advisories are rarely tied to the actual developmental state of an individual plant or to the conditions measured at its roots. At the same time, low-cost microcontrollers, cloud telemetry and lightweight object-detection models have made it technically feasible to monitor a small plot continuously and to extract structured information from ordinary smartphone photographs [8]–[10]. Reviews of the field nevertheless report that the high cost of sensors, analytics platforms and expertise remains a barrier for smallholders [8], which motivates a deliberately affordable and interpretable design.

1.2 Overall Project Context

NAIMIRIS (project R26-IT-103) is a group research project that aims to support economically viable Scotch Bonnet cultivation. The group architecture separates pre-cultivation land and investment decision support from cultivation-stage guidance and other specialised modules; each module is the individual research responsibility of a different student. This thesis examines only one of them: **Technology-Assisted A-to-Z Farming Guidance**, which supports the grower from germination to fruit maturity. The pre-analysis component can supply a registered farm and plot profile, and a separate component addresses pest and disease detection. Neither of these is assessed here, and Section Section 1.9 defines the boundary precisely.

1.3 Individual Component Context

The component receives two complementary kinds of evidence. The first is a stream of environmental measurements from an ESP32-based field node: air temperature, relative humidity, soil moisture, soil temperature and soil electrical conductivity (EC). The second is a set of smartphone photographs that the farmer captures at defined crop checkpoints, which stage-specific YOLO object detectors convert into structured observations such as the number of visible seedlings, flowers or fruits. A stage-aware backend combines both streams with the plant's recorded age and history, applies reliability checks, and returns a monitoring status, alerts and practical, low-risk recommendations to a mobile application that also preserves a growth history for the farmer [8], [11].

The proposal that began this research (March 2026) described a four-week early-stage monitor based on YOLOv8 localisation followed by a ResNet-18 stage classifier. The later project record (the PP2 presentation and the August 2026 implementation report) extends coverage to five developmental stages and replaces the single classifier with task-specific detectors. The scope change is substantive rather than cosmetic: guidance that stops after nursery establishment cannot help a farmer manage irrigation, EC-related nutrition, reproductive-stage heat stress and harvest timing across a productive crop cycle. Table 1.1 summarises the revised scope.

Table 1.1. Revised five-stage component scope (PP2 and implementation specification).

Stage	Primary observation	Checkpoint / approach	Decision-support role
Germination	Root-zone conditions	Continuous sensor monitoring	Moisture and thermal suitability
Seedling establishment	Seedling emergence	IoT plus YOLOv8n emergence detection (reported in PP2)	Emergence evidence and establishment guidance
Vegetative growth and branching	Leaf and canopy development	IoT; Week 7 visual checkpoint (specified, not independently verified)	Development monitoring; imaging pathway to be confirmed
Flowering and pollination	Visible flowers and young fruits	Weeks 10 and 11; YOLO flowering detector	Flowering evidence and microclimate warnings
Fruiting and ripening	Unripe and ripening fruit instances	Weeks 16 and 20; YOLO11s fruit detector	Visible crop load, ripening progress, harvest planning

1.4 Problem Statement

A sensor-only display reports the condition of the environment but not whether the plant is visibly progressing; a photo-only detector reports visible plant structures but cannot explain the root-zone conditions that accompany them. The smallholder is left to translate both sources, manually and without stage-specific reference values, into decisions about water, shade and inputs. The problem can be decomposed into four sub-problems:

- **P1: Stage ambiguity.** Crop age is only an approximate indicator of physiological stage, so thresholds that ignore the confirmed stage give misleading advice.
- **P2: Measurement validity.** Low-cost sensor values are only meaningful once their physical basis is known; a raw soil-moisture reading is not a percentage of field capacity, and EC values from different measurement bases are not interchangeable [12], [13].
- **P3: Evidence alignment.** A photograph and a sensor reading describe the same plant only if they are close enough in time, location and quality to be interpreted together.
- **P4: Actionability and trust.** A number or a count is not advice. Farmers need short, specific, appropriately cautious guidance that states what evidence it rests on and what it cannot conclude.

The research problem is therefore to develop an affordable, interpretable and traceable method for linking stage-specific visual evidence with temporally aligned environmental observations, so that *Nai Miris* growers receive actionable monitoring guidance instead of isolated numerical readings or organ counts.

1.5 Research Gap

The gap addressed is an integration and decision-design gap, not a claim that no comparable system exists. The literature reviewed in Chapter 2 establishes that YOLO-family detectors can recognise chilli growth stages and maturity under defined conditions [14], [15], and that IoT platforms can deliver continuous remote telemetry [8], [11]. What is far less developed, and what farm-facing action depends on, is the layer between these two: how to confirm the stage, qualify sensor validity, align image and sensor evidence, interpret deviations with persistence rather than at a single sample, and communicate the result with calibrated caution. Table 1.2 states the gap as four specific, testable shortfalls that map onto the objectives.

Table 1.2. Research gaps and the objectives that address them.

Gap	What is missing in the reviewed evidence	Addressed by
G1	Chilli vision studies focus on a single task (growth stage or maturity) in controlled or hydroponic conditions and are not tied to concurrent root-zone measurements.	O1, O3, O4
G2	IoT studies demonstrate telemetry and threshold alarms, but rarely document calibration basis, sensor validity checks or stage-specific profiles for a named crop.	O2, O4
G3	Neither stream is routinely aligned in time or checked for quality before being interpreted together, so conflicting evidence is not handled explicitly.	O4
G4	Reported systems seldom specify reproducible acceptance tests for detectors, sensors, alert logic and farmer comprehension within one traceable protocol.	O5

1.6 Aim and Objectives

Aim. To design, implement and evaluate a practical AI- and IoT-supported A-to-Z cultivation guidance component for *Nai Miris* that transforms stage-specific visual features and environmental monitoring into interpretable farmer decisions.

The aim is decomposed into five objectives. Each has a verifiable deliverable, and Table 1.3 links objectives to research questions and to the chapter in which each is addressed.

- **O1.** Build a crop-cycle workflow covering five developmental stages with checkpoint-based image capture.
- **O2.** Capture and quality-check ESP32 environmental measurements and persist them for recent and historical assessment.
- **O3.** Apply stage-appropriate YOLO detectors for seedling emergence, flowering and fruit/ripening tasks.
- **O4.** Develop explainable rules that join detector evidence with calibrated, stage-specific environmental information.
- **O5.** Specify reproducible model, sensor, system and stakeholder tests that establish reliability and usefulness.

Table 1.3. Traceability of objectives to research questions, deliverables and thesis chapters.

Objective	Research question	Primary deliverable	Design (Ch. 4)	Evaluation (Ch. 3 protocol)
O1	RQ1	Stage state machine; checkpoint schedule; capture protocol	Sections 4.5, 4.7	Stage-routing tests; usability
O2	RQ1, RQ4	Validated telemetry pipeline; aggregate store	Sections 4.4, 4.9	Sensor error, coverage and drift study
O3	RQ2	Three stage-specific detectors with documented class maps	Section 4.7	Precision, recall, mAP, count error
O4	RQ1, RQ3	Rule engine, alert lifecycle, recommendation template	Sections 4.6, 4.8	Boundary tests; expert rubric; false-alert replay
O5	RQ4	Test cases, traceability matrix, evidence register	Section 4.12	Protocol in Sections 3.7 to 3.9

1.7 Research Questions and Hypotheses

- **RQ1.** How can sensor readings and image observations be aligned and interpreted across the *Nai Miris* growth cycle?
- **RQ2.** What visual information can YOLOv8 and YOLO11 models reliably extract at selected checkpoints?
- **RQ3.** How can a rules engine explain environmental risks without asserting causality that the evidence does not support?

- **RQ4.** Which performance, usability and validity measures are necessary before field deployment?

Because this is a design-science study, the central proposition is a design hypothesis rather than a causal agronomic claim. It is stated in three testable parts, to be examined in the evaluation chapter:

- **H1 (decision quality).** Stage-conditioned guidance that fuses image and sensor evidence is rated by agronomic reviewers as more appropriate, and as making fewer unsupported causal claims, than image-only or sensor-only guidance on the same cases.
- **H2 (alert quality).** Persistence-and-recovery alert logic produces fewer false or duplicate alerts than single-sample thresholding when both are replayed on the same recorded telemetry, without missing sustained excursions.
- **H3 (usability).** Target users can correctly identify the current status and the recommended action from the guidance screen, and rate the application above the conventional System Usability Scale benchmark [16], [17].

None of these hypotheses asserts a yield or profit benefit. Whether guided farms outperform unguided farms requires a controlled, multi-season field trial that lies beyond the scope of this work.

1.8 Significance and Expected Contributions

The work contributes at three levels. At the **practical** level it offers a low-cost, smartphone-and-ESP32 workflow that a smallholder can operate, with advice phrased as low-risk, reversible steps. At the **engineering** level it specifies a stage-aware pipeline in which every recommendation is traceable to versioned thresholds, a model version, a time-aligned sensor window and a data-quality state. At the **methodological** level it provides a reproducible verification protocol that separates specification-derived acceptance tests from field evidence, and that is explicit about plant-disjoint evaluation, calibration basis and threats to validity. The principal research artefact is therefore the traceable multi-input interpretation workflow, not a new neural-network backbone and not a validated agronomic prediction model.

1.9 Scope, Assumptions and Limitations

In scope: crop-cycle registration; environmental sensing and validation; checkpoint image analysis; stage-based alerts; human-readable cultivation recommendations; dashboards; and a reproducible evaluation design.

Out of scope: pest identification and disease classification (a separate team component); automatic pesticide, fertiliser or irrigation control; land-suitability planning; market pricing and financial return modelling; and any claim about whole-plant yield.

Assumptions and limitations. (a) Sensor thresholds are provisional engineering seed values that require local agronomic review. (b) Detector counts refer to objects visible in a submitted photograph and are not whole-plant ground truth. (c) Crop response, generalisation across farms and fertiliser dosing are not experimentally established by the available evidence. (d) The system is advisory; it never actuates equipment. (e) Environmental risk indicators are discussed only as guidance context and are never used as pest or disease classification.

1.10 Evidence Basis and Status of the Work

The thesis draws on four project artefacts: the March 2026 individual proposal, the October 2026 individual thesis template, the August 2026 A-to-Z implementation report and the PP2 presentation. The proposal describes a preliminary four-week YOLOv8n/ResNet-18 concept. The PP2 record reports the five-stage workflow, ESP32/Firebase monitoring, YOLOv8n seedling detection, a YOLO-based flowering detector and a YOLO11s fruit detector. The implementation report supplies the detailed technical specification and the proposed evaluation protocol. Neither document contains a fully auditable test set, confusion matrices, field yield comparisons or sensor-calibration logs, and the thesis therefore does not report such results. To keep every statement honest, content is classified into three maturity levels, shown in Table 1.4 and used consistently in later chapters.

Table 1.4. Evidence-maturity classification used throughout the thesis.

Level	Meaning	Examples in this thesis

Level	Meaning	Examples in this thesis
Reported (PP2)	Functionality reported as implemented in the project prototype; independent benchmark evidence not yet attached	Five-stage workflow; ESP32 to Firebase updates; YOLOv8n seedling, flowering and YOLO11s fruit detectors; stage-aware dashboard
Specified	Design or protocol taken from the implementation report; to be reconciled with the repository	Threshold seed profiles; persistence rule; ±10-minute alignment window; API endpoints; Week 7 imaging checkpoint
To be validated	Requires data or experiments not yet available	Detector mAP and count error; sensor calibration and drift; alert false-positive rate; farmer usability; agronomic utility

1.11 Individual Contribution

Savinda R.M.D (IT22124630) is responsible for the cultivation-guidance component: ESP32 sensor-monitoring integration, the crop-stage monitoring workflow, growth-stage image analysis, alert rules and cultivation recommendations, and the associated mobile views. Other components supply complementary capabilities and are not claimed as personal work. Integration interfaces, such as shared plot identifiers, do not imply shared ownership of research methods.

1.12 Structure of the Thesis

Chapter 2 reviews the agronomic, sensing, computer-vision, data-fusion and human-factors literature and the existing systems, and synthesises the research gap into a conceptual framework. Chapter 3 presents the design-science methodology, data and materials, procedures, evaluation metrics and statistical analysis plan, threats to validity, ethics and reproducibility measures. Chapter 4 specifies requirements, architecture, the detailed design of each subsystem, the data and interface model, security design and the verification strategy. Implementation, results and discussion are reported in subsequent chapters outside the scope of this document.

2 LITERATURE REVIEW AND RELATED WORK

This chapter reviews the evidence that informs the design of a stage-aware guidance system for *Nai Miris*. It covers crop physiology and phenology, low-cost environmental sensing, IoT architectures and alarm design, vision-based crop detection, multimodal fusion and explainable decision support, and human factors. The chapter closes with a critical synthesis that converts the findings into explicit design implications and a conceptual framework.

2.1 Review Method and Scope

A focused narrative and design-oriented review was conducted. The starting point was the reference set of the original March 2026 proposal. It was extended by targeted retrieval of peer-reviewed and standards-based sources for five themes: (i) *Capsicum* environmental physiology; (ii) soil and air sensing and its calibration; (iii) IoT telemetry and alarm management; (iv) object detection for crops, with emphasis on chilli; and (v) data fusion, explainability, usability and evaluation methodology. Sources were selected when they met at least one of the following criteria: they study *Capsicum* or a closely comparable horticultural crop; they document a method or metric that the thesis relies on; or they provide authoritative specifications for a tool used in the design (for example the Ultralytics documentation for the deployed YOLO versions). Recent work was preferred for fast-moving topics such as object detection, and foundational publications were retained for established methods such as statistical agreement analysis.

This is not a PRISMA-style systematic review: search strings, screening counts and exclusion logs were not recorded, and the thesis does not claim exhaustive coverage. Each source is therefore appraised for the function it can support, the conditions of its experiment and the transferability of its findings to smallholder *Nai Miris* cultivation in Sri Lanka. Table 2.1 summarises the evidence families and their principal limitation.

Table 2.1. Evidence families, representative sources, relevance and transfer limitations.

Evidence family	Representative sources	Relevance to this thesis	Transfer limitation

Evidence family	Representative sources	Relevance to this thesis	Transfer limitation
Chilli production and policy context	[1]–[3]	Crop importance, smallholder constraints, phenology	Policy and general agronomy; not sensor or algorithm evidence
Capsicum physiology	[4], [18]–[20]	Stage-sensitive responses to temperature and salinity	Growth-chamber, altitude-gradient and *C. annuum* studies; no local thresholds
Sensing and calibration	[12], [13], [21]–[23]	Meaning and validity of moisture, EC, temperature, humidity readings	Principles are general; probe-specific calibration is required
IoT and alarm design	[8], [9], [11], [24]	Telemetry pipelines, alarm persistence and deadband	Reviews are broad; many studies are not crop-specific or are on large farms
Crop vision	[14], [15], [25]–[27]	Stage and maturity detection, small-object and occlusion issues	Hydroponic or orchard settings; different cultivars and cameras
Dataset shift and evaluation	[28]–[32]	Why plant-disjoint, local test data are needed; metric definitions	Benchmarks are not *Nai Miris* results
Fusion and explainability	[33]–[36]	Late fusion, conflicting evidence, interpretable rules	General frameworks; not agricultural advisory studies
Human factors	[16], [17], [37]–[39]	Acceptance, usability measurement	Instruments need local-language adaptation

2.2 Chilli Cultivation in Sri Lanka and the Nai Miris Crop

Policy analysis for the Agriculture Sector Modernization Project identifies chilli as a crop where productivity gains and improved agronomic practice could reduce import dependence and raise farm incomes [1], and the Horticultural Crop Research and Development Institute documents the capsicum varieties and recommended practices for Sri Lankan conditions [2]. Habanero-type *C. chinense* is a perennial species cultivated as an annual in most production systems. Its life cycle is conventionally divided into germination, seedling, vegetative growth, flowering, fruit set and fruit maturation, with a prolonged harvest period as fruits ripen successively [3]. The crop is therefore managed over many months, which makes it a good candidate for stage-aware monitoring and also means that errors in stage assumptions propagate through every subsequent recommendation.

Two features of the context shape the design. First, holdings are small and capital is limited, so any monitoring solution must rely on inexpensive hardware and the farmer's existing smartphone. Second, the existing advisory landscape is dominated

by general information services rather than plant-level, evidence-based guidance; the existing systems are reviewed in Section 2.9.

2.3 Capsicum Phenology and Environmental Physiology

2.3.1 Phenology and the limits of age-based staging

Phenology is the ordered development of a plant through recognisable stages. Calendar age is an operational proxy for stage but an imperfect one, because cultivar, transplanting date, microclimate and management all shift the timing. Image-based growth-stage recognition has been proposed precisely because arbitrary time divisions can disagree with the true state of the plant. In the chilli context, Schneider et al. trained YOLOv8 models to classify three stages (growing, flowering, fruiting) in a hydroponic grower, using four camera systems and both bird's-eye and side views, and found that models trained on both views generalised better [14]. Two lessons follow for this work: stage evidence should come from images as well as dates, and camera view is an explicit source of variation that the capture protocol must control.

2.3.2 Temperature responses

Temperature is the best-documented environmental driver of *C. chinense* performance. Garruña-Hernández et al. grew Habanero plants in chambers with diurnal maximum air temperatures of 30, 35 and 40 °C and concluded that the thermal optimum is 30 to 35 °C when expressed as leaf temperature rather than air temperature; at 40 °C air temperature, raised stomatal conductance and transpiration lowered leaf temperature by about 5 °C [4]. Jaimez et al. compared sites with mean temperatures of 19, 24 and 28 °C and reported delayed flowering and fruit production at the cooler sites, with many aborted flowers and small fruits, and an increase in flower number as temperature rose between roughly 26 and 30 °C [18]. Work on bell pepper (*C. annuum*) shows that sensitivity to elevated temperature differs between flower developmental stages and organs [19].

> **Design implication.** Because the response depends on leaf rather than air temperature, and on the developmental stage at which the excursion occurs, a single air-temperature limit cannot be asserted as a universal safety threshold. The air-temperature seed bands used in this thesis (Table 4.6) are therefore treated as conservative, reviewable engineering defaults, and alerts are worded as "possible heat stress" rather than diagnosed damage.

2.3.3 Water, salinity and nutrition

Pepper is classified as moderately sensitive to soil salinity; the widely used salt-tolerance assessment places the yield threshold of pepper at about 1.5 dS m^{-1} of saturation-extract electrical conductivity (ECe), with yield declining linearly above it [20]. Experiments with greenhouse pepper hybrids confirm that the effect of NaCl salinity depends on the growth stage at which it is applied [40]. Field studies of hot pepper also show that fertiliser regime changes growth and yield, but the optimum is site- and source-specific [41]. Together these studies justify monitoring EC as an indicator, and they argue against turning an EC value into a fertiliser-dose instruction without local calibration: the thesis restricts EC-related advice to verification and expert consultation.

2.4 Low-Cost Sensing: Validity, Calibration and Measurement Basis

2.4.1 Soil moisture

Most low-cost soil-moisture probes infer volumetric water content from the dielectric permittivity of the soil, or from a related proxy such as capacitance or resistance. The relation between the raw output and volumetric water content is soil-specific, as the classic dielectric calibration of Topp et al. and subsequent reviews of measurement methods make clear [13], [21]. A raw analogue-to-digital value, or even a manufacturer percentage, is not a percentage of field capacity; it becomes one only after a documented calibration against gravimetrically determined water content in the actual growing medium, and after a field-capacity reference has been defined. This is a central validity constraint of the thesis: values are reported as "% FC" only when that calibration exists.

2.4.2 Electrical conductivity

Soil salinity is conventionally expressed as the electrical conductivity of a saturated-paste extract (ECe), standardised to 25 °C. In-situ probes instead measure bulk or pore-water conductivity, which varies with water content and temperature and is not interchangeable with ECe or with the EC of a nutrient solution [12]. Any EC threshold must therefore record its measurement basis and reference temperature, and a conversion or empirical calibration is needed before in-situ readings are compared with ECe-based limits such as those in [20].

2.4.3 Air temperature and humidity; the ESP32 measurement chain

The DHT22 (AM2302) digital sensor widely used in low-cost nodes is specified by its manufacturer with an accuracy on the order of ±0.5 °C for temperature and ±2 to ±5 % for relative humidity, with a slow response [22]; exposure to direct sun and self-heating can add error unless the sensor is shielded and ventilated. On the ESP32, analogue inputs are read by a 12-bit converter whose transfer curve is documented as non-linear, particularly towards the ends of the range, and the second ADC unit cannot be used while Wi-Fi is active, so the choice of input pins and the use of the vendor calibration scheme affect accuracy [23]. These hardware properties motivate three design measures in Section 4.4: median aggregation with a coverage count, an explicit calibration record per probe, and an acquisition design that avoids the Wi-Fi-conflicting ADC unit.

2.5 IoT Architectures, Connectivity and Alarm Management

Systematic reviews describe agricultural IoT as layered: a perception layer of sensors, a network layer, and application and analytics layers that provide decisions to users [8], [9]. Ahmed et al. demonstrate such a layered platform with LoRaWAN for large Chilean farms and show that real-time monitoring can drive alarms and irrigation decisions, while noting that rural connectivity is a design constraint [11]. For a smallholder plot with a household Wi-Fi or mobile hotspot, a direct Wi-Fi link to a managed real-time database is simpler and cheaper than a LoRa gateway, whereas LoRa becomes attractive where Wi-Fi is unavailable; this trade-off is revisited in Section 4.3. Firebase's Realtime Database provides synchronised, low-latency state suitable for live dashboards [42], and a Python FastAPI service provides

schema-validated REST endpoints following the resource-oriented style described by Fielding [43], [44].

A less discussed but crucial element is alarm design. The process-industry alarm-management standard ANSI/ISA-18.2 identifies nuisance alarms and alarm floods as a major source of operator error and recommends practices such as on-delay (requiring a condition to persist before alarming), deadband or off-delay (requiring sustained return before clearing), de-duplication and rationalisation of every alarm to a defined operator action [24]. Although agricultural monitoring is a much lower-stakes setting, the same human-factors logic applies: a farmer who receives repeated spurious warnings will stop reading them. The persistence-and-recovery rules specified in Section 4.6 are direct adaptations of these principles.

2.6 Computer Vision for Crop Monitoring

2.6.1 Object detection and the YOLO family

Object detectors predict, for each object in an image, a class label, a confidence score and a bounding box. The YOLO ("You Only Look Once") approach reframed detection as a single regression pass over the image, trading some localisation precision for real-time speed [45], and has since evolved through many versions [46]. YOLOv8 adopted an anchor-free, decoupled-head design and is distributed in nano to extra-large scales; YOLOv8n has about 3.2 million parameters [47]. YOLO11 refined the backbone and neck, and the small variant (YOLO11s) has about 9.4 million parameters according to its documentation [48]. These figures describe model complexity on the COCO benchmark [32]; they are not evidence of accuracy on *Nai Miris* images, which must be measured on held-out local data.

2.6.2 Chilli and fruit detection studies

Wang et al. proposed YOLO-ALW, a YOLOv8n-based chilli maturity detector that targets occlusion, small targets and fruit–foliage colour similarity, and reported high detection scores on their own dataset [15]. The broader fruit-detection literature, from early deep-network systems for sweet pepper and rock melon [26] to orchard detection [27] and comprehensive reviews of deep learning for fruit detection and yield estimation [25], consistently identifies occlusion, illumination change, variable

fruit size and limited labelled data as the principal difficulties. Surveys of deep learning in agriculture add that data availability and the gap between curated datasets and real field conditions remain open challenges [49].

2.6.3 Dataset shift, augmentation and small datasets

Torralba and Efros showed that visual datasets carry systematic biases that reduce performance when a model is applied to a different collection [28]. In plant imaging, Mohanty et al. reported that a classifier trained on controlled-background leaf images performed substantially worse on images from other sources [29]. Data augmentation (flips, colour jitter, mosaics and similar transforms) can improve robustness but does not create independent biological samples [50]: if augmented copies of the same plant appear in both training and test partitions, measured accuracy is inflated. These findings underpin three procedures in Chapter 3: collecting local imagery under a defined protocol, splitting by plant or session before augmentation, and reporting results per class and per acquisition condition.

2.6.4 Evaluation metrics

Detection quality is conventionally summarised by precision, recall, average precision (AP) and mean AP (mAP) at an intersection-over-union (IoU) threshold, with the PASCAL VOC and COCO protocols defining how matches and averages are computed [30]–[32]. For counting tasks, which are the practical output here, detection metrics should be complemented by count-agreement statistics such as mean absolute error and bias against manual annotation. The Ultralytics documentation defines the metrics reported by the training tool [51]; the thesis adopts these definitions and reports them with the matching confidence and IoU settings.

2.7 Sensor–Vision Fusion and Explainable Decision Support

Multisensor data fusion can operate at the data, feature or decision level, and the literature emphasises that real inputs are imperfect, conflicting, correlated and misaligned in time [33]. Multimodal machine-learning surveys identify alignment and representation of heterogeneous inputs as core challenges [34]. In this component the two modalities are very different (a low-rate numeric series and a

sparse set of images), the amount of expert-labelled advisory data is small, and every output must be understandable by a farmer. Decision-level (late) fusion through explicit rules, applied after each stream has been validated and aligned, is therefore the appropriate choice. This position is consistent with arguments that, when stakes involve real costs to users, interpretable models should be preferred to post-hoc explanations of black boxes [35], and with the call for rigorous, application-grounded evaluation of interpretability [36]. It does not exclude learned models in future work once enough expert-labelled cases exist.

2.8 Human Factors and Technology Acceptance

Adoption of agricultural technology depends on perceived usefulness and perceived ease of use [37], and usability is defined by ISO 9241-11 as the effectiveness, efficiency and satisfaction with which specified users achieve specified goals in a specified context [38]. The System Usability Scale provides a quick, widely used instrument, and empirical work offers adjective-based interpretation of its scores [16], [17]. Nielsen's heuristics supply design principles for screen design such as visibility of system status and match between the system and the real world [39]. For smallholders, these principles translate into short messages, icon-plus-text status, local-language readiness (Sinhala and Tamil) and tolerance of poor connectivity.

2.9 Existing Systems, Products and Techniques

National and regional initiatives such as the IWMI smart-farming programme [6] and the Govi Sahaya agriculture platform [7] provide information, advisory content and community functions for Sri Lankan farmers, and a range of commercial IoT dashboards provide sensor telemetry. These are retained as contextual comparisons. No claim is made that they categorically lack any feature: a controlled feature audit would be needed before asserting a competitive advantage. Table 2.2 compares approach types by their evidence source, strengths and limitations relative to the target of plant-level, stage-aware guidance.

Table 2.2. Conceptual comparison of approaches (qualitative; not an empirical product benchmark).

Approach	Main evidence	Strength	Typical limitation relative to the target

Approach	Main evidence	Strength	Typical limitation relative to the target
Manual crop inspection	Farmer's visual judgement	Cheap and context-rich	Time-intensive; inconsistent documentation; expertise-dependent
General advisory platforms	Regional information and community knowledge	Broad reach; low cost	Not tied to the plant's measured state or stage
IoT-only dashboard	Sensor telemetry	Continuous observation	No organ-level evidence; thresholds often generic
Image-only YOLO tool	Photographic detections	Quantifies visible objects	No root-zone context; viewpoint-dependent counts
Proposed stage-aware fusion	Stage + validated sensors + YOLO detections	Explainable, traceable guidance	Requires local calibration and field validation

2.10 Critical Synthesis and Conceptual Framework

The reviewed evidence supports four conclusions. First, *C. chinense* responses to temperature, water and salinity are stage-dependent and are documented mainly in controlled or foreign conditions, so local thresholds are provisional. Second, low-cost sensor values are meaningful only with a stated calibration basis. Third, chilli vision models perform well on their own datasets but are vulnerable to dataset shift, occlusion and viewpoint, so counts must be treated as view-limited observations. Fourth, interpretable, persistence-aware, evidence-attached rules are a defensible way to combine the two streams. Table 2.3 converts these conclusions into design decisions that are developed in Chapter 4.

Table 2.3. Design implications derived from the literature.

Finding	Source family	Design decision	Where specified
Responses are stage- and organ-dependent; leaf, not air, temperature governs optimum	Physiology	Versioned stage profiles; "possible stress" wording; no universal limit	Section 4.6.1
Raw sensor output is not a physical quantity without calibration	Sensing	Calibration record; basis field; suppress diagnoses when invalid	Section 4.4
EC bases are not interchangeable	Sensing	Basis-specific EC profiles; advice limited to verification	Section 4.6.1
Nuisance alarms erode trust	Alarm management	On-delay, deadband, de-duplication, cooldown	Section 4.6
Detector accuracy does not transfer between datasets	Vision	Local data; plant-disjoint split; per-condition reporting	Chapter 3
Counts depend on view and occlusion	Vision	Capture protocol; "visible count" semantics; recapture on poor quality	Section 4.7

Finding	Source family	Design decision	Where specified
Fusion inputs are misaligned and imperfect	Fusion	Time-window alignment; evidence-quality flags; conflict rules	Section 4.8
Interpretability matters when decisions have costs	Explainability	Rule engine with decision trace; versioned rule IDs	Section 4.8
Usability and acceptance drive adoption	Human factors	Short guidance cards; SUS and comprehension tests	Section 4.10

The resulting conceptual sequence is shown in Figure 2.1: registered crop-cycle context and two evidence streams are each validated, stage-specific detection and stage-aware interpretation follow, evidence is aligned and fused by rules, and the advisory output is returned to the farmer, whose acknowledgements and corrections are written to an audit trail.

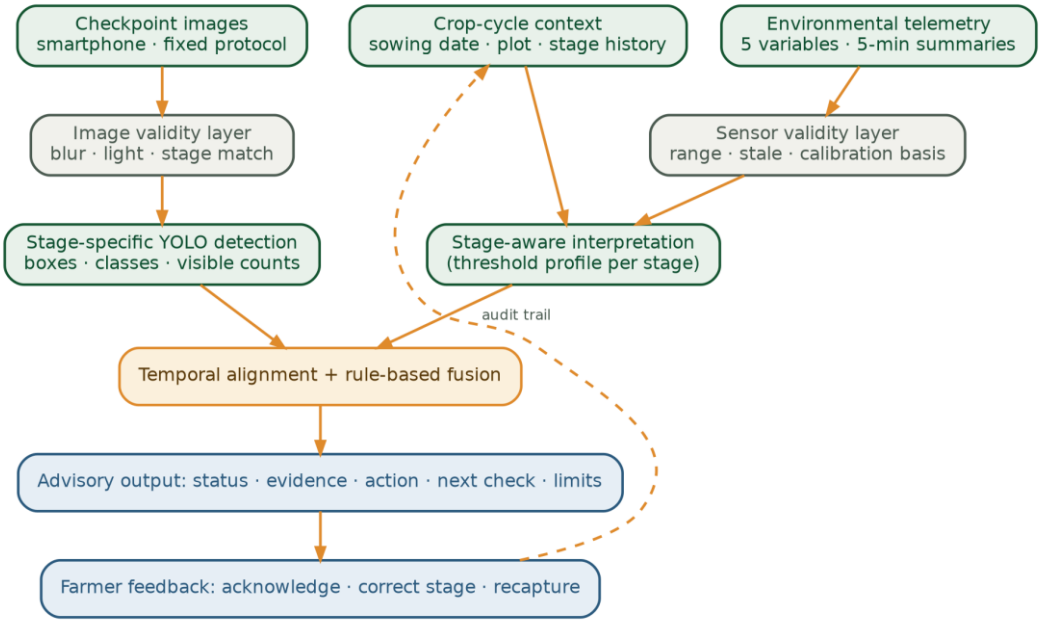


Figure 2.1. Conceptual framework of stage-aware, multi-evidence cultivation guidance.

2.11 Chapter Summary

The literature supports the feasibility of each building block but reveals an integration gap at the point where validated sensor evidence, stage context and image observations must be turned into cautious, auditable advice. It also shows that the main risks are not algorithmic novelty but measurement validity, dataset shift, alert

fatigue and comprehension. The next chapter describes the research methodology designed to build and evaluate the artefact with those risks controlled.

3 RESEARCH METHODOLOGY

This chapter describes how the artefact is designed, built and evaluated. It states the research paradigm and design, the process and its mapping to established design-science guidance, the stakeholders and data, the procedures for each subsystem, the metrics and statistical plan, the threats to validity, and the ethical, security and reproducibility measures. Procedures that depend on data not yet collected are written as protocols; results are deliberately excluded.

3.1 Research Design

The study adopts a pragmatic stance: the aim is to produce a working solution to a practical decision problem and to evaluate it with the methods best suited to each claim (quantitative metrics for detectors and sensors, expert rubric for advice quality, and usability instruments for farmers). **Design Science Research (DSR)** is the governing methodology because the investigation creates and evaluates an information-technology artefact, the stage-aware guidance workflow, for a defined problem [52], [53]. The lifecycle comprises problem identification, requirements elicitation, system and model design, prototype development, iterative technical testing and targeted stakeholder evaluation. Table 3.1 shows how the seven guidelines of Hevner et al. are satisfied or, where evidence is not yet available, how they will be satisfied.

Table 3.1. Mapping of DSR guidelines to this research.

Guideline [52]	How it is addressed	Status
1. Design as an artefact	A stage-aware, multi-evidence guidance workflow with ESP32, Firebase, FastAPI, YOLO detectors and a mobile interface	Reported (PP2) and specified
2. Problem relevance	Smallholder *Nai Miris* growers lack plant-level, stage-specific advice (Section Section 1.4)	Argued with sources
3. Design evaluation	Detector, sensor, alert, fusion, usability and expert-rubric evaluations (Sections Section 3.7 to Section 3.8)	Protocol defined; results pending
4. Research contributions	Traceable fusion workflow; validity-aware sensing rules; reproducible verification protocol (Section Section 1.8)	Specified
5. Research rigour	Plant-disjoint splits, calibration records, pre-declared metrics, versioned artefacts	Specified
6. Design as a search process	Iterations from the four-week ResNet-18 concept to five-stage task-specific detectors (Section Section 4.3.2)	Documented

Guideline [52]	How it is addressed	Status
7. Communication	Thesis, project presentations, repository and evidence register	In progress

3.2 Research Process and Project Phases

The six activities of the DSR process model [53] are mapped to the thesis and to the project timeline in Figure 3.1. Evaluation findings feed back into design: for example, a high false-alert rate on replayed telemetry would lead to retuning the persistence parameters, and poor count agreement on a given view would lead to revising the capture protocol.

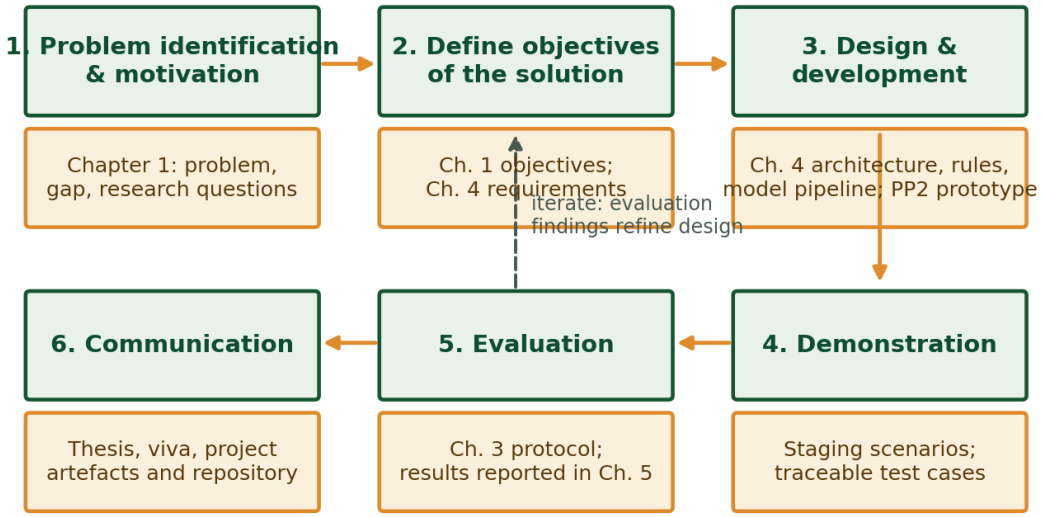


Figure 3.1. Research process: DSR activities mapped to thesis chapters and project artefacts.

The practical development followed the phases summarised in Table 3.2. The March proposal defined a four-phase approach; the later implementation work expanded it to cover sensor stabilisation, stage rules, guidance, image checkpoints, fusion and evaluation.

Table 3.2. Project phases, principal outputs and evidence status.

Phase	Principal output	Evidence status
P0 Proposal (March 2026)	Problem definition; four-week YOLOv8n/ResNet-18 concept; FR1–FR9 and NFR1–NFR5	Documented (historical design)

Phase	Principal output	Evidence status
P1 Sensor stabilisation	ESP32 acquisition of five variables; Firebase updates; payload schema	Reported (PP2); calibration logs pending
P2 Stage rules and guidance	Five-stage profiles; five-band classification; persistence; dashboard	Reported / specified
P3 Image checkpoints	Capture protocol; YOLOv8n seedling, flowering and YOLO11s fruit detectors	Reported (PP2); independent test pending
P4 Fusion and recommendations	Time alignment; rule engine; recommendation template	Specified; partial prototype
P5 Evaluation	Execution of the protocol in this chapter	To be performed

3.3 Stakeholders and Requirements Elicitation

Requirements were derived from four sources: the proposal's functional and non-functional requirements; the features reported in PP2; the implementation specification; and the constraints identified in the literature (Chapter 2). Because no formal farmer survey has yet been documented, stakeholder needs in Table 3.3 are stated as working assumptions, and structured validation with real users is part of the evaluation protocol rather than a completed step. Requirements are prioritised with the MoSCoW scheme (Must, Should, Could, Won't for this release) and every functional requirement is given an acceptance test.

Table 3.3. Stakeholders, needs and planned involvement.

Stakeholder	Primary need (working assumption)	Planned involvement
Smallholder grower	Clear stage status, simple low-risk actions, low cost, works with poor connectivity	Task-based usability session; SUS; comprehension questions
Agricultural extension officer / agronomist	Trustworthy thresholds and cautious wording; ability to audit advice	Review of threshold profiles; blinded rating of advice cases
Research team / supervisors	Traceability, reproducibility, correct scope boundaries	Design reviews; evidence register
Other NAIMIRIS components	Stable plot and plant identifiers; authenticated interface	Interface contract and integration tests

3.4 Data, Participants and Study Materials

The study materials are smartphone crop photographs, bounding-box annotations, timestamped environmental telemetry, crop-cycle metadata and expert-reviewed

recommendation cases. Table 3.4 lists each stream with its required fields, quality controls and current evidence status.

Table 3.4. Study data streams, required fields, quality controls and current status.

Data stream	Required fields	Quality controls	Status from project artefacts
IoT telemetry	Plant ID, device ID, UTC time, variable, unit, value, quality flag	Range, stale and duplicate checks; calibration reference	ESP32/Firebase monitoring reported (PP2)
Seedling images	Image, plant ID, acquisition date, annotation	Blur and lighting checks; plant-separated split	YOLOv8n detector reported
Flower images	Boxes, visible-flower labels, view	Consistent view; ambiguous-bud rule	YOLO flowering detector reported (version unspecified)
Fruit images	Unripe/ripening boxes, model version	Occlusion and maturation-ambiguity flags	YOLO11s detector reported
Expert-labelled cases	Case, suggested action, reviewer rating	Blinding, rubric, inter-rater agreement	Protocol proposed

3.4.1 Annotation guideline and class definitions

Before any further labelling, an annotation guideline (version 1.0) fixes the class definitions, because ambiguity at class boundaries is the main source of label noise in maturity and flowering detection. The guideline specifies, for each class, the visual criterion for inclusion, the minimum visible fraction below which an object is not labelled, and the treatment of overlapping and partly occluded objects. The reference scheme is: *seedling* (an emerged seedling with cotyledons visible above the medium); *flower* (an open corolla visible in the image, with buds either excluded or placed in a separate class decided once and recorded); *young fruit* (swollen ovary after petal drop); *unripe fruit* (fruit with the pre-ripening colour of the cultivar); and *ripening fruit* (fruit showing visible colour break towards the ripe colour). Two annotators label an overlapping subset, disagreements are adjudicated, and agreement is reported with Cohen's kappa [54].

3.4.2 Dataset provenance and independence

The project training record for the fruit detector reports approximately 118 initial images. This count is treated as a source-dataset description to be confirmed against the final dataset manifest; it is not an independent test-set size. Augmented copies do not constitute independent biological samples [50], and images from public or

hydroponic sources are used only as proxies and flagged as such. For every dataset the manifest records image hash, source, licence, plant ID, acquisition session, camera, view and label version. The effective sample size for inference is the number of independent plants or sessions, not the number of images. Field participation, permissions and sample sizes remain to be documented before quantitative validation.

3.5 Procedures

3.5.1 Sensor acquisition, calibration and preprocessing

The crop cycle and device identifiers are registered; the node samples the five signals and transmits five-minute summaries via Firebase. Each summary carries the median of the valid samples, the number of valid samples and a coverage ratio, and is stamped with UTC acquisition time. Values outside physical range, repeated identical readings that indicate a frozen sensor, and messages older than a freshness window are flagged and excluded from rule evaluation. Missing data are reported as "unavailable" and are never treated as zero. Calibration follows a written protocol:

- **Soil moisture.** Multi-point calibration of the probe output x against volumetric water content determined gravimetrically in the actual growing medium, fitting a documented function $\theta = f(x)$; field capacity (θFC) is then determined for the same medium, and the reported value is $\%FC = 100\ \theta / \theta FC$ [13], [21].
- **Electrical conductivity.** Verification against a standard conductivity solution, conversion of readings to 25 °C with $EC25 = ECT / [1 + \alpha (T - 25)]$, with $\alpha \approx 0.02\ °C^{-1}$ to be fitted, and an explicit record of the measurement basis (in-situ bulk, pore water or saturated-paste extract) [12].
- **Air temperature and humidity.** Side-by-side comparison with a reference thermo-hygrometer, with a radiation-shielded, ventilated mounting; agreement is analysed with the Bland–Altman method [22], [55].
- **Drift and coverage.** Repeat checks at defined intervals; coverage and missing-rate statistics reported per device.

3.5.2 Stage confirmation and band classification

The active stage is taken from the recorded crop-cycle history, with explicit farmer confirmation of stage changes when the date-based schedule and the image evidence disagree. For each valid parameter the system selects the single active profile (species, cultivation method, stage, sensor and measurement basis) and classifies the aggregate into one of five bands: critical-low, low, target, high or critical-high. A deviation raises an alert only when it persists for n consecutive valid aggregates (the specification proposes $n = 3$, about 15 minutes), and an alert is resolved only after m consecutive in-band aggregates ($m = 3$). The persistence rule follows the on-delay and off-delay practice of alarm management [24], and n and m are configuration values that are tuned empirically, not experimentally optimal constants.

3.5.3 Checkpoint image capture and inference

At project checkpoints (Week 7 where specified, Weeks 10 and 11, and Weeks 16 and 20) the farmer is guided to photograph the plant at a repeatable distance, angle and lighting. Preprocessing validates file type and size, corrects orientation and screens for gross quality problems. The image is then routed to the detector matching the active stage: YOLOv8n for seedling emergence, the flowering detector for visible reproductive structures, and YOLO11s for unripe and ripening fruit. Non-maximum suppression and a documented confidence threshold, chosen on the validation set, produce the visible counts. A low-confidence or heavily occluded image prompts recapture instead of a confident statement.

3.5.4 Temporal alignment, fusion and recommendation

The photograph timestamp is matched to the nearest valid sensor aggregate within a configurable window (the specification suggests ±10 minutes). If none exists, the fused conclusion is withheld and the image result is shown with a stale-sensor notice. The stage, visual features, sensor level and direction, trend, and evidence quality are combined by rules. Where several causes are plausible, the output uses qualified wording such as “possible heat stress” and advises inspection and sensor confirmation. The recommendation, its evidence identifiers, the threshold and rule versions, and the farmer’s acknowledgement are stored for audit.

3.6 Model Development Protocol

Each detector is fine-tuned from publicly available pretrained weights on the project dataset using the Ultralytics framework [47], [48]. To keep the work reproducible, the protocol fixes the following before the test set is touched: the data split and its random seed; the augmentation policy (geometric and photometric transforms applied to the training partition only); the training budget and early-stopping rule on the validation set; the image size; and the selection of the confidence threshold from the validation precision–recall behaviour. The weights, configuration file and framework version are archived with a checksum. The test partition is evaluated once, after all choices are frozen, and any later change creates a new model version with a new test report. Hyper-parameters actually used are reported in the results chapter and are not asserted here.

3.7 Evaluation Design and Metrics

Evaluation is organised by subsystem so that an error can be attributed to its source. Figure 3.2 shows the data flow for the vision evaluation, in which splitting by plant precedes augmentation and the test partition is sealed.

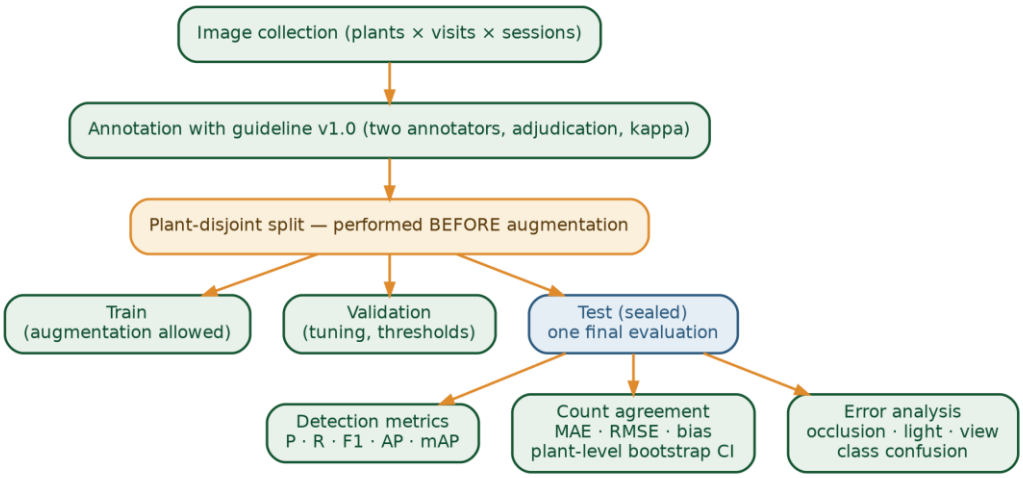


Figure 3.2. Vision evaluation protocol with plant-disjoint, sealed test partition.

3.7.1 Detection and counting metrics

A predicted box is a true positive when its IoU with an unmatched ground-truth box of the same class exceeds the chosen threshold; otherwise it is a false positive, and unmatched ground-truth boxes are false negatives [30], [31].

$$\mathrm{IoU} = |B_p \cap B_g| \,/\, |B_p \cup B_g| \tag{3.1}$$

$$P = TP / (TP + FP) \qquad R = TP / (TP + FN) \qquad F1 = 2PR / (P + R) \tag{3.2}$$

$$AP_c = \int_0^1 p_c(r)\, dr \qquad mAP = (1/C) \sum_c AP_c \tag{3.3}$$

Here p_c(r) is the precision of class c as a function of recall r, and C is the number of classes; mAP is reported at IoU 0.50 and averaged over IoU thresholds 0.50 to 0.95 in steps of 0.05. Because the product output is a count, agreement with manual counts is evaluated on quality-approved comparable photographs, where $\hat{y}$ is the predicted and y the manually annotated count for image i of n:

$$\mathrm{MAE} = (1/n) \sum |\hat{y}_i - y_i| \tag{3.4a}$$

$$\mathrm{RMSE} = \sqrt{[\ (1/n) \sum (\hat{y}_i - y_i)^2\ ]} \tag{3.4b}$$

$$\mathrm{Bias} = (1/n) \sum (\hat{y}_i - y_i) \tag{3.4c}$$

$$\rho = N_{ripening} / (N_{unripe} + N_{ripening}) \times 100\ \% \tag{3.5}$$

The visible ripening ratio ρ is defined only when N_unripe + N_ripening > 0; it is reported as “not estimable” when no fruit is detected, never as 0 %. Official mAP values in vendor documentation are on the COCO dataset and must not be read as the performance of the *Nai Miris* weights [32], [48].

3.7.2 Sensor, alert, fusion and usability metrics

Sensor accuracy is expressed as mean absolute error, bias and limits of agreement against reference instruments, together with coverage, missing-rate and drift [55]. Alert behaviour is measured by the false-alert rate per sensor-day, duplicate-notification rate, detection delay for injected sustained excursions, and time to resolution, comparing persistence-based logic with single-sample thresholding on identical replayed telemetry. Temporal fusion is measured by the fraction of images with a valid sensor snapshot and the distribution of snapshot age. Usability combines task completion, time on task, comprehension questions and the System Usability Scale: SUS = 2.5 × [$\Sigma(s_i - 1)$ over the five odd-numbered items + $\Sigma(5 - s_i)$ over the five even-numbered items], where each item score s_i is on a 1–5 scale [16].

Table 3.5. Evaluation plan: dimension, protocol, reference evidence and proposed acceptance criterion.

Dimension	Metric and protocol	Reference evidence required	Proposed acceptance criterion
Image detection	P, R, F1, AP50, mAP50–95 per class on sealed plant-disjoint test set	Independent ground-truth boxes; split manifest	Reported with confidence intervals; no fixed target before data exist
Visible counts	MAE, RMSE, bias per class and view	Blinded human annotation	Bias and limits reported; recapture rate reported
IoT calibration	Absolute error, drift, coverage, missing rate; Bland–Altman	Reference instruments; calibration sheets	Limits of agreement inside the tolerance agreed with the agronomist
Alert logic	Boundary cases; false/duplicate alerts; delay; recovery	Deterministic scenarios; replayed logs	All boundary cases pass; H2 comparison reported
Temporal fusion	Valid-snapshot fraction; snapshot age	Timestamped images and readings	Unmatched cases always withhold a confident conclusion
Latency	p50 / p95 image-to-guidance time on intended hardware	Timed staging runs	p95 ≤ 3 s (proposal target; to be demonstrated)
Usability	Task completion; comprehension; SUS	Consenting farmers and advisers	Mean SUS compared with the 68 benchmark (H3)
Agronomic utility	Appropriateness, clarity, urgency, unsupported-claim rate	Blinded expert rubric; inter-rater kappa	Kappa ≥ 0.61 (substantial) before rubric results are used

3.8 Statistical Analysis Plan

Descriptive plots show sensor trajectories and the time spent outside locally calibrated target bands. Detection and count errors are reported separately for each class, site and camera condition. Because photographs of the same plant are correlated, confidence intervals are obtained by a **plant-level (cluster) bootstrap**, resampling plants and not images [56]; proportions such as recapture rate use Wilson score intervals [57]. Sensor agreement uses Bland–Altman bias and 95 % limits of agreement [55]. Inter-annotator and inter-rater agreement is quantified with Cohen's kappa and interpreted with the conventional bands [54].

For H1, expert ratings of image-only, sensor-only and fused guidance on the same cases are compared with a paired non-parametric test on ordinal ratings, with unsupported-causal-claim counts compared descriptively. For H2, the two alert policies are replayed on identical telemetry and compared by false-alert and duplicate counts per sensor-day, with the sustained-excursion detection rate reported alongside

to show that fewer alerts are not bought by missing real events. For H3, mean SUS is reported with a confidence interval and compared with the benchmark of 68 [17]; with a small number of participants these results are reported as exploratory. Formal agronomic benefit requires a controlled prospective trial and cannot be inferred from simulations or replayed logs.

3.9 Threats to Validity

Table 3.6. Threats to validity and planned mitigations.

Type	Threat	Mitigation
Internal	Same-plant images in train and test partitions inflate accuracy	Split by plant or session before augmentation; sealed test set
Internal	Threshold, persistence or confidence tuned on test data	Tune on validation only; freeze before test; version every change
Construct	Visible counts mistaken for whole-plant counts; %FC or ECe mislabelled	"Visible count" semantics; calibration basis recorded and displayed
Construct	Ripening colour varies with exposure and white balance	Capture protocol; report errors by lighting condition
External	Single farm, cultivar or camera; hydroponic or foreign proxy images	Multi-site, multi-device sampling; flag proxy data; report per condition
External	Seed thresholds taken from literature on other conditions	Treat as provisional; agronomist review before field use
Conclusion	Small samples and correlated observations	Cluster bootstrap; exploratory labelling of small-n results
Conclusion	Rater bias in expert assessment	Blinding to guidance type; pre-declared rubric; kappa

3.10 Ethics, Privacy, Security and Responsible AI

Informed consent is obtained before collecting farm photographs, dates, device telemetry or interview data, and participants can withdraw. Personally identifying metadata are minimised, precise location is restricted, and retention and deletion periods are defined, in line with the principles of Sri Lanka's Personal Data Protection Act [58]. Security measures follow the OWASP API Security guidance [59]: authenticated routes, object-level authorisation (a farmer can read only their own plants), HTTPS, role-scoped database rules and protected device credentials. Responsible-AI measures include uncertainty and sensor-quality notices in every recommendation, warnings against unverified fertilisation or irrigation changes, and an explicit statement that the system is advisory and never an autonomous controller.

Every recommendation must be explainable and reversible. Use of generative AI in preparing this document is disclosed in Appendix B.

3.11 Reproducibility and Research Artefacts

Training datasets and label maps are versioned; source licences, split strategy and random seeds are recorded; model checkpoints are stored with hashes; and the framework, firmware and telemetry-schema versions are pinned. A minimal anonymised scenario pack, representative inference outputs, calibration worksheets and stage-rule unit tests are archived. Re-running an experiment must recover the same model and test data instead of retraining on silently changing cloud resources. Screenshots and logs are added to the appendices when available.

3.12 Project Risks and Contingency

Table 3.7. Principal project risks.

Risk	Likelihood / impact	Mitigation or contingency
Insufficient independent plant images for a meaningful test set	High / High	Prioritise multi-session capture; report effective sample size; use proxy images only for pre-training
Sensor drift or probe failure in the field	Medium / High	Heartbeat monitoring; sensor-quality flags; scheduled re-verification; spare probes
Connectivity loss	High / Medium	Local buffering and idempotent retry; stale-data notices that are distinct from stress alerts
Seed thresholds disagree with agronomist review	Medium / Medium	Versioned profiles; update without code change; expert sign-off before field use
Model version mismatch between training and deployment	Low / High	Model registry with checksums; version stored with every analysis
Low farmer adoption or comprehension	Medium / High	Early usability testing; short guidance cards; local-language readiness

3.13 Chapter Summary

The methodology combines design-science rigour with a measurement-first view of the data: every number must have a calibration basis, every detector a plant-disjoint evaluation, and every recommendation a trace. The following chapter applies this methodology to specify the product and solution design.

4 PRODUCT AND SOLUTION DESIGN

This chapter specifies the guidance component: its design principles, requirements, architecture and the detailed design of each subsystem, followed by the data model, interfaces, user interface, security design and the verification strategy. Design elements are labelled with the evidence levels of Table 1.4. Where the implementation report specifies a feature that the PP2 record does not demonstrate, the text says so and the item must be reconciled with the repository before final submission.

4.1 Design Overview and Principles

Five principles, derived from the literature synthesis in Table 2.3, guide every design decision:

- **D1. Evidence before assertion.** A recommendation states the stage, the evidence (sensor windows, detections, versions) and its limits; unsupported causal claims are not made.
- **D2. Validity before interpretation.** Readings and images are screened first; invalid, stale or uncalibrated inputs suppress diagnoses and are reported as such.
- **D3. Stage awareness.** Targets, checkpoints, detectors and advice are selected from the confirmed crop stage, never from a single global profile.
- **D4. Graceful degradation and safe defaults.** When evidence is missing, conflicting or poor, the system asks for recapture or inspection and does not guess; device or connectivity faults are never presented as crop stress.
- **D5. Farmer-centred, auditable and advisory.** Output is short, plain-language and reversible, every action is logged, and nothing is actuated automatically.

4.2 Requirements

Requirements were derived from the proposal's FR1–FR9 and NFR1–NFR5, the PP2 features, the implementation specification and the literature. The revised scope replaces the exclusively four-week stage classifier with a five-stage system and task-specific detectors. Every requirement is expressed as a testable behaviour, prioritised with MoSCoW, and linked to an objective and to acceptance tests in Section 4.12.

Table 4.1 lists functional requirements and Table 4.2 the non-functional requirements, organised by ISO/IEC 25010 quality characteristics [60].

Table 4.1. Functional requirements with priority, objective and acceptance evidence.

ID	Functional requirement	Pri.	Obj.	Acceptance evidence
FR-01	Create and maintain plant, plot and crop-cycle records including stage history	Must	O1	Stage timeline and edit audit (TC-06 to TC-11, TC-28)
FR-02	Collect five environmental variables through ESP32 and Firebase with timestamps and quality flags	Must	O2	Validated sample stream (TC-01 to TC-05)
FR-03	Display stage-relevant targets and trend history	Must	O2, O4	Dashboard and time-series checks (TC-29)
FR-04	Support checkpoint image capture with quality screening	Must	O1, O3	Valid/invalid image cases (TC-16, TC-24)
FR-05	Run seedling, flowering and fruit detection and route by stage	Must	O3	Class/version mapping and inference fixtures (TC-08 to TC-10, TC-17)
FR-06	Match image evidence with the nearest valid sensor window	Must	O4	Time-window and missing-data tests (TC-18)
FR-07	Generate explainable warnings with severity, persistence and resolution	Must	O4	Rule boundary and lifecycle tests (TC-11 to TC-14, TC-19 to TC-21)
FR-08	Display current stage, detections, counts, evidence and next steps	Must	O1, O4	Usability and integration tests (TC-29, TC-30)
FR-09	Store evidence, recommended actions and acknowledgements	Must	O4, O5	Database audit and API tests (TC-27, TC-30)
FR-10	Suppress agronomic diagnosis when calibration basis, freshness or quality is invalid	Must	O2, O4	TC-03 to TC-05
FR-11	Distinguish device or connectivity faults from crop-stress alerts	Must	O2	TC-15, TC-26
FR-12	Allow farmer-confirmed stage correction while keeping the machine suggestion	Should	O1	TC-28
FR-13	Provide history charts and exportable evidence for expert review	Could	O5	Export test (future)

Table 4.2. Non-functional requirements, quality characteristics and design targets.

ID	ISO 25010 characteristic	Design target and qualification
NFR-01	Performance efficiency	Measure p50 and p95 image-to-guidance latency; the proposal target of 3 s is a target and is not yet demonstrated
NFR-02	Reliability	Track ingestion uptime and coverage; the proposal figure of 99.9 % is an aspiration, not a measured result
NFR-03	Usability	Short messages, icon plus text, local-language readiness; evaluated by SUS and comprehension tasks
NFR-04	Reliability (recoverability)	Device-side buffering, idempotent retry and safe stale-data behaviour
NFR-05	Security	Authenticated routes, HTTPS, object-level authorisation, least-privilege database rules

ID	ISO 25010 characteristic	Design target and qualification
NFR-06	Maintainability / traceability	Persist model, profile and rule versions with every analysis; update profiles without redeploying code
NFR-07	Portability / compatibility	Plot and plant identifiers and timezone-safe timestamps compatible with other NAIMIRIS components
NFR-08	Security (confidentiality)	Data minimisation; location precision restricted; retention and deletion policy
NFR-09	Cost efficiency (project constraint)	Low-cost node and smartphone-only capture; bill of materials to be recorded

4.3 Architecture Within the Complete Solution

The architecture has six logical parts: sensing hardware; Firebase transport and live state; stage and telemetry analysis; checkpoint image processing; multimodal recommendation; and the mobile user interface (Figure 4.1). ESP32 observations carry structured event identifiers and timestamps. The FastAPI layer validates and summarises data, selects stage profiles and stores alerts. Smartphone images are uploaded separately for model inference. The fusion engine combines the confirmed stage, the image result and the most recent acceptable sensor window, and the mobile application presents actions and history. Long-term relational tables are specified as the target architecture and are not described as verified deployed infrastructure without repository or database evidence.

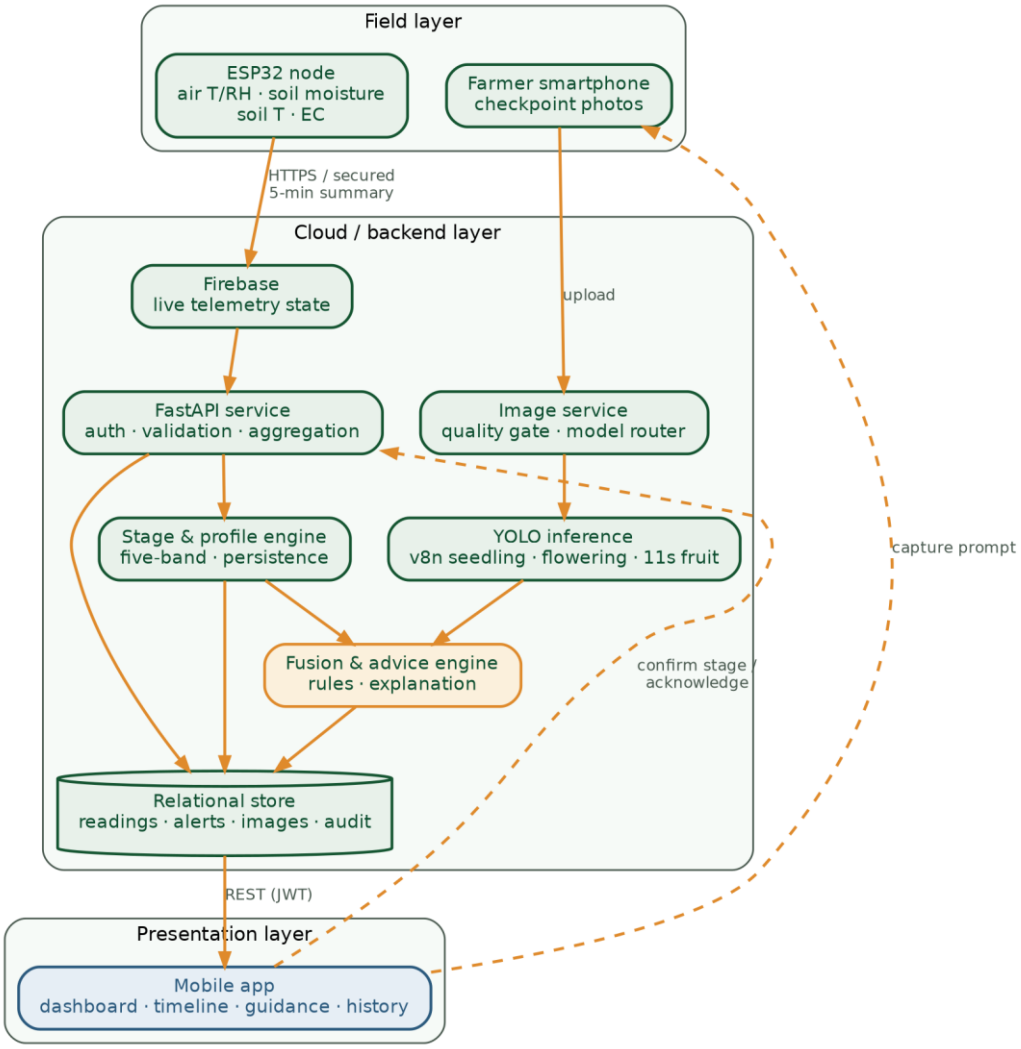


Figure 4.1. Logical architecture of the A-to-Z guidance module (field, backend and presentation layers).

System boundary. The pre-analysis component may provide a land or farm registration profile; other components stay separate. This component performs no pest classification, financial return calculation or market operation. Integration requires common plot and plant identifiers, authenticated requests, timezone-safe timestamps and explicit indication of stale data.

4.3.1 Technology choices

Table 4.3. Technology choices and rationale.

Layer	Choice	Rationale and caveat

Layer	Choice	Rationale and caveat
Field node	ESP32 microcontroller	Integrated Wi-Fi, low cost, wide community support; ADC non-linearity and ADC/Wi-Fi pin constraints managed in firmware [23]
Transport and live state	Firebase Realtime Database	Low-latency synchronised state; supported by PP2; dependence on a managed service and on connectivity [42]
Backend	FastAPI (Python)	Schema validation, asynchronous endpoints and a natural home for the Ultralytics inference stack [43]
Vision	Ultralytics YOLOv8n / YOLO11s	Real-time detectors with documented training and export tools; small models suit modest hardware [47], [48]
Long-term store	Relational database (target)	Strong constraints for audit entities; to be confirmed against the repository
Client	Mobile application	Smartphone is already owned by the farmer and provides the camera

4.3.2 Design alternatives and decisions

Table 4.4. Architecture trade-offs, options considered and reasoned selection.

Design question	Options considered	Selection and caveat
Vision architecture	One universal model vs task-specific detectors	Task-specific detectors reduce label ambiguity across very different objects (seedling, flower, fruit); more models to maintain and version
Sensor transport	Direct broker (e.g., MQTT) vs Firebase relay	Firebase supported by PP2 and simple to integrate; dependency and connectivity cost accepted
Connectivity	Wi-Fi vs LoRaWAN	Wi-Fi for plots with a household network; LoRa gateway recommended where Wi-Fi is absent [11]
Inference location	Phone or edge vs backend	Backend inference eases model updates and keeps weights private; latency and offline trade-off accepted
Alerts	One-shot threshold vs persistent five-band logic	Persistence and recovery reduce noise [24]; parameters tuned empirically
Advice engine	Black-box model vs explicit rules	Auditable rules until enough expert-labelled cases exist [35]
EC interpretation	One generic EC band vs measurement-basis profiles	Basis-specific profiles prevent invalid comparison [12]

Design evolution. The proposal suggested YOLOv8 localisation followed by a ResNet-18 weekly-stage classifier. The project evidence subsequently supports a different implementation: YOLOv8n seedling detection, a YOLO flowering detector and YOLO11s fruit detection. The ResNet-18 workflow is therefore retained only as historical design context and is not reported as the deployed pipeline. Likewise, the vegetative-stage image analysis is distinguished from the models shown as implemented in PP2.

4.4 Sensing Subsystem Design

The field node measures five variables. Table 4.5 states, for each, the quantity, the calibration requirement and the validity rule; sensor models are to be confirmed against the project's bill of materials. The node samples at a short interval, computes a robust five-minute summary and publishes it, together with device metadata, over an authenticated channel.

Table 4.5. Measurement chain: variable, calibration requirement and validity rule.

Variable	Calibration / basis requirement	Validity rule (examples)
Soil moisture	Probe-specific calibration to volumetric content in the actual medium; field-capacity reference before reporting %FC	Reject below zero or above physical maximum; flag frozen values; report "raw" until calibrated
Soil temperature	Offset check against reference thermometer; installation depth recorded	Plausible range and maximum rate of change
Soil EC	Standard-solution check; temperature compensation to 25 °C; measurement basis recorded	Compare with ECe-based limits only when basis is compatible
Air temperature	Comparison with reference; radiation shield and ventilation	Plausible range; rate-of-change limit
Relative humidity	Comparison with reference; sensor warm-up and response time respected	0–100 % bounds; consistency with temperature

Acquisition design. Analogue probes are connected to inputs of the ADC unit that remains available while Wi-Fi is active, and the vendor calibration scheme is applied [23]. Within each five-minute window the node takes repeated samples, discards values outside range, and publishes the median, the number of valid samples, the coverage ratio and a quality flag. A heartbeat message is published independently of readings so that the backend can distinguish a silent node from a stable plant. The illustrative message structure is shown below and must be reconciled with the actual firmware schema.

```
{
  "message_id": "esp32-01-20261010T0930Z-0187",
  "device_id": "esp32-01", "plant_id": "P-001", "cycle_id": "C-2026-04",
  "window_start_utc": "2026-10-10T09:25:00Z",
  "firmware": "0.9.2", "calibration_ref": "CAL-2026-09-30",
  "values": {
    "soil_moisture": {"median": 68.2, "unit": "pct_fc",  "basis": "calibrated", "n_valid": 58,
"coverage": 0.97},
    "soil_temp":     {"median": 24.6, "unit": "degC",    "basis": "probe",      "n_valid": 60,
"coverage": 1.00},
    "air_temp":      {"median": 27.9, "unit": "degC",    "basis": "shielded",   "n_valid": 60,
"coverage": 1.00},
    "air_rh":        {"median": 61.0, "unit": "pct",     "basis": "shielded",   "n_valid": 59,
"coverage": 0.98},
    "soil_ec":       {"median": 0.92, "unit": "dS_per_m","basis": "insitu_25C", "n_valid": 55,
"coverage": 0.92}
  }
```

```
}
```

The message identifier makes ingestion idempotent: a duplicate message creates no second reading. Messages are accepted only from registered devices with valid credentials, and a payload whose measurement basis is absent is stored but excluded from threshold comparison (TC-04).

4.5 Stage Management Subsystem

Stage is the key context for every other subsystem. The crop cycle starts at sowing and moves through five stages, with the checkpoint weeks of Figure 4.2 used as a schedule rather than as biological constants. The system keeps three pieces of information: the schedule-derived stage, any image evidence of stage, and the confirmed stage actually used. A transition occurs when the schedule and evidence agree, or when the farmer confirms it; when they disagree the farmer is asked to confirm, and the machine suggestion and the farmer's decision are both stored with source and time (FR-12). Rules that require a particular stage use only the confirmed stage.

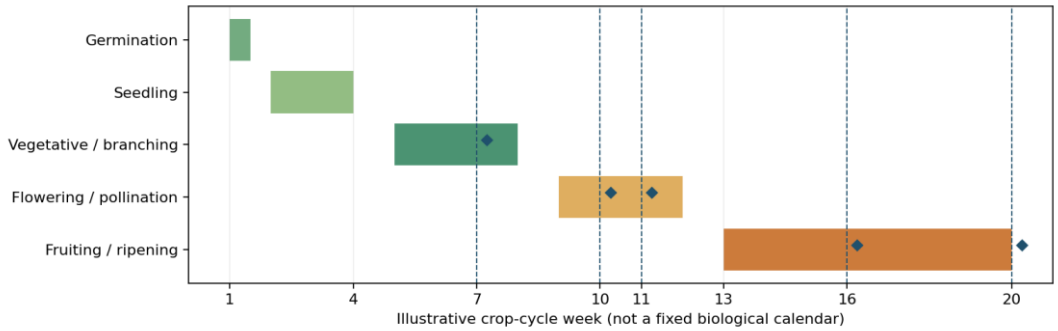


Figure 4.2. Stage durations and image checkpoint schedule (schematic; planned week boundaries, not validated biological constants).

Image checkpoints are scheduled per stage: Week 7 for the vegetative stage where specified, Weeks 10 and 11 for flowering, and Weeks 16 and 20 for fruiting. Continuous sensor monitoring runs in all stages, and the seedling stage additionally uses the emergence detector. Pairs of images (Weeks 10 and 11) are comparable only if the capture protocol was followed, which is why TC-22 forbids reporting a flower-drop rate from mismatched views.

4.6 Telemetry Analysis and Alert Engine

4.6.1 Stage profiles and the five-band model

A threshold profile is a versioned record keyed by species, cultivation method, stage, sensor and measurement basis, holding the band limits and the persistence parameters *n* and *m*. For every valid five-minute aggregate exactly one active profile is selected. Four limits define five bands: critical-low, low, target, high and critical-high. The implementation specification lists six limit fields (critical_low, low, optimal_low, optimal_high, high, critical_high); here the four outer limits define the bands and the intermediate fields are treated as optional early-warning margins, to be reconciled with the repository. Table 4.6 gives the target ranges transcribed from the specification.

Table 4.6. Prototype reference target ranges transcribed from the implementation specification (unvalidated seed values).

Stage	Moisture (%FC)	Soil temp (°C)	RH (%)	Air temp (°C)	Soil ECe (dS/m)
Germination	70–85	26–30	70–90	24–30	0.30–0.80
Seedling	60–75	22–27	55–75	21–28	0.50–1.00
Vegetative	60–80	20–28	50–70	20–30	0.70–1.30
Flowering	65–80	20–27	50–70	20–29	0.80–1.40
Fruiting	65–80	20–28	50–70	20–30	0.80–1.50

These are engineering seed values pending localised agronomic calibration, not transferable crop standards. The specification gives target ranges only; the critical limits must be set during agronomist review and are not invented here.

The seed values are consistent in magnitude with the literature but must be read critically. The upper EC target of 1.5 dS/m coincides with the ECe yield threshold reported for pepper [20], which is expressed on a saturation-extract basis that an in-situ probe does not measure. The air-temperature ceilings of 28 to 30 °C are below the 30 to 35 °C leaf-temperature optimum reported for Habanero [4]; since leaf temperature can be several degrees below air temperature in hot conditions, the air-temperature ceilings are conservative and are a priority for expert review. Figure 4.3 visualises the moisture targets.

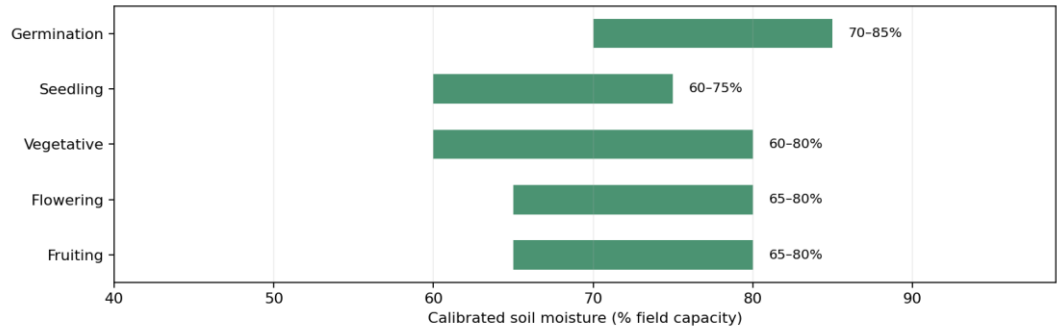


Figure 4.3. Provisional moisture target bands by stage (design thresholds from the implementation guide; not validated field findings).

A soil EC value cannot be read as an exact nitrogen concentration, and %FC cannot be inferred from an arbitrary raw ADC value. Rule evaluation therefore suppresses agronomic diagnoses when the calibration basis, timestamp or sensor quality is invalid (FR-10).

4.6.2 Classification algorithm

Band assignment is deterministic and inclusive at the target edges, so a value exactly on a boundary is classified as target (TC-11):

```
def classify(v, p):                       # p = active threshold profile
    if v is None or not valid(v):  return UNAVAILABLE
    if v <  p.critical_low:        return CRITICAL_LOW
    if v <  p.optimal_low:         return LOW
    if v <= p.optimal_high:        return TARGET      # inclusive edges
    if v <= p.critical_high:       return HIGH
    return CRITICAL_HIGH


def update(state, band, sample_valid):
    if not sample_valid:           return SUPPRESSED    # never counted as in-band
    if band != TARGET:  state.dev += 1; state.ok = 0
    else:               state.ok += 1;  state.dev = 0
    if state.dev >= n and not state.active:  open_alert(state, band)
    if state.ok  >= m and state.active:      resolve_alert(state)
```

4.6.3 Persistence, deduplication and alert lifecycle

An out-of-band sample first puts the parameter in a watch state that is not shown to the farmer. The alert opens only after *n* consecutive valid deviating aggregates (specified as three, about 15 minutes) and resolves after *m* consecutive in-band aggregates (three), as the lifecycle in Figure 4.4 shows. While an alert is open, further abnormal samples update the same record and do not create duplicate notifications; a cooldown applies after resolution. Invalid or stale data move the parameter to a suppressed state instead of counting as in-band or out-of-band. These parameters are configuration values and are not experimentally optimised.

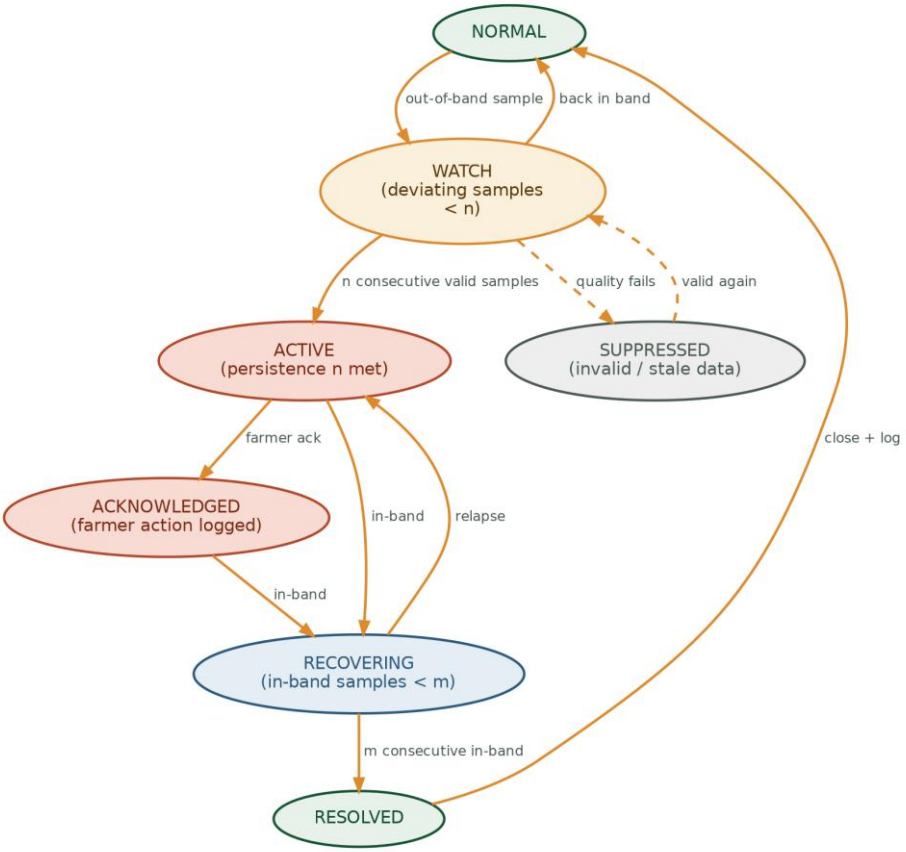


Figure 4.4. Alert lifecycle with persistence, recovery and data-validity suppression.

Figure 4.5 illustrates the effect on a synthetic signal that was generated for illustration only: an isolated spike does not alert, a sustained drop does, and resolution requires a sustained return.

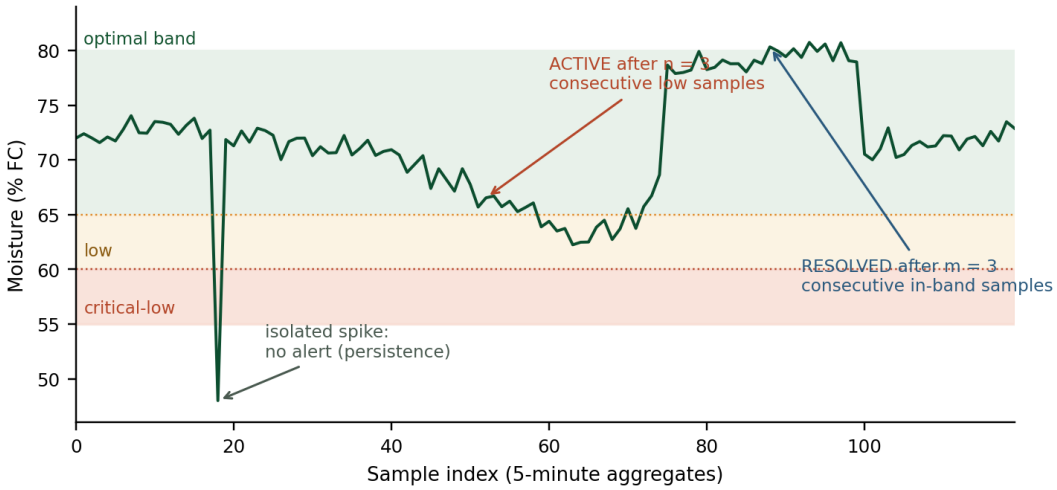


Figure 4.5. Synthetic illustration of persistence-based alerting (not measured data).

4.6.4 Severity and notification policy

Table 4.7. Severity policy (advisory only).

Condition	Severity	Farmer-facing treatment
Single or fewer than n deviating samples	None (internal watch)	Not shown; logged
Low or high band, persistent	Advisory	Dashboard card; suggested low-risk check
Critical band, persistent	Warning	Push notification; action checklist; recheck time
Persistent deviation in a sensitive stage (flowering, fruiting) with corroborating image evidence	Warning with priority	Push notification; recommends inspection and recapture
Silent device or stale data	Device notice (not crop stress)	Connectivity message; last valid timestamp shown

4.7 Vision Subsystem Design

4.7.1 Capture protocol and quality gate

Count comparability depends on the capture protocol: a repeatable distance, angle (side or bird's-eye view, fixed per checkpoint), daylight lighting and a plain framing guide shown on the capture screen. Schneider et al. found that camera view changes generalisation [14], so the view is recorded as metadata. The server-side quality gate checks file type and size, orientation, sharpness (for example the variance of the Laplacian), exposure range and minimum resolution. Numerical limits are calibrated on pilot images and stored in configuration, not hard-coded. Failures return a specific reason and a recapture request (Figure 4.6).

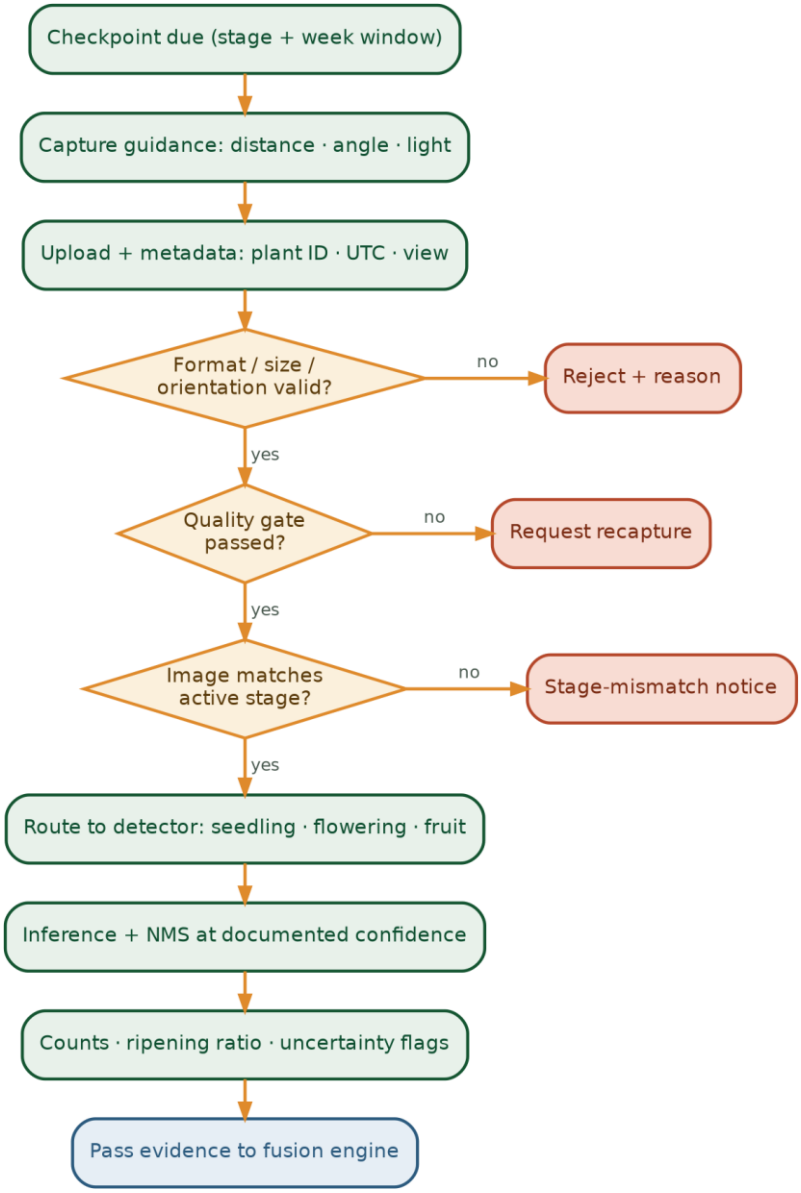


Figure 4.6. Checkpoint image-processing pipeline with validation, quality and stage gates.

4.7.2 Detector configuration

Table 4.8. Stage-specific detectors.

Stage / checkpoint	Model	Output (classes to be confirmed against the checkpoint label map)	Evidence status
Seedling establishment	YOLOv8n	Emerged-seedling boxes, confidence, visible count	Reported (PP2)
Flowering, Weeks 10 and 11	YOLO flowering detector (version unspecified)	Visible flowers and young fruits where trained and labels confirmed	Reported (PP2); version to be recorded

Stage / checkpoint	Model	Output (classes to be confirmed against the checkpoint label map)	Evidence status
Fruiting, Weeks 16 and 20	YOLO11s	Unripe and ripening boxes, confidence, per-image counts	Reported (PP2); about 118 initial images in training record
Vegetative, Week 7	To be confirmed	Canopy or leaf-development observation	Specified; not independently verified

YOLOv8 and YOLO11 are described as distinct model families, not synonyms [47], [48]. Inference applies non-maximum suppression and the validation-selected confidence threshold. The ripening proportion follows Equation 3.5: $\rho = N_{ripening} / (N_{unripe} + N_{ripening})$, reported as "not estimable" when no fruit is detected. For illustration only, 7 ripening and 13 unripe visible detections would give 35 % visible ripening; this arithmetic example is not a measured result. Ripening appearance varies with illumination, so ρ is an observational signal and not a guarantee of chemical maturity. Each analysis stores the file hash, capture time, plant ID, view, model version, boxes, quality flags and whether manual review was requested.

4.7.3 Dataset curation and model lifecycle

Images enter a versioned dataset manifest with provenance and licence; annotation follows guideline v1.0 (Section Section 3.4.1); splits are by plant or session before augmentation; models are trained under the protocol in Section Section 3.6 and registered with checkpoint hash, label map, configuration and test report. The service loads a model only through the registry, so an analysis can always be traced to the exact weights that produced it. Inference results with low confidence or heavy occlusion are marked uncertain and never interpreted as absence of fruit (TC-24).

4.8 Evidence Fusion and Recommendation Engine

4.8.1 Temporal alignment and evidence quality

Alignment selects, for each image, the sensor aggregate with the smallest absolute time difference inside the configured window (specified as ±10 minutes) among aggregates with acceptable quality. The result carries an evidence-quality descriptor with four fields: snapshot age, sensor validity, image quality and detector confidence. A fused conclusion is permitted only when all four meet minimum conditions;

otherwise the engine returns the individual results with a note on what is missing (TC-18, TC-24). Quality is expressed as categories (good, limited, poor) and not as a pseudo-probability, because no calibrated probabilistic model exists to justify a numeric confidence.

4.8.2 Rule representation and decision trace

Rules are data, not code: each has an identifier and version, the stage it applies to, required evidence from each stream, the cautious interpretation, the advised low-risk steps, an urgency and a next-check interval. An illustrative rule follows.

```
rule_id: R-FLW-HEAT-01
version: 1.0
stage: flowering
requires:
  image:   { model: flowering, quality: good }
  sensors: { air_temp: [HIGH, CRITICAL_HIGH], air_rh: [LOW], persistent: true,
             validity: valid, max_age_min: 10 }
interpretation: "Possible reproductive-stage heat stress (not a diagnosis)"
advice: [check_shade_ventilation, inspect_flowers_closely, keep_irrigation_stable]
urgency: warning
next_check_min: 120
limits: "Visible flower count depends on view and occlusion"
```

When a rule fires, the engine stores a decision trace: the rule ID and version, the profile version, the identifiers of the sensor aggregates and image analysis used, the evidence-quality descriptor and the generated text. This trace is what makes advice auditable (NFR-06) and is the basis for expert rating in the evaluation.

4.8.3 Rule catalogue and conflict handling

Table 4.9. Explainable rule examples (cautious interpretations, not confirmed diagnoses).

Visual evidence and sensors	Cautious interpretation	Farmer-facing guidance example
Flowering + persistent hot, dry air	Possible reproductive environmental stress	Check shade and ventilation; inspect flowers; keep irrigation stable
Fruiting + unstable moisture	Potential fruit-quality risk; not a diagnosis	Inspect irrigation uniformity and visible fruit symptoms
Visible fruit ripening + valid conditions	Harvest-readiness cue for visible fruit only	Recheck colour in daylight; plan selective harvest
Abnormal sensor + low data quality	Device or calibration fault possible	Inspect the sensor before changing crop inputs
Low moisture + high air temperature + low RH	Possible drought and heat load	Inspect soil and irrigation; consider shade; recheck after 2 h (TC-19)
High moisture + high RH	Possible waterlogging and poor ventilation	Review drainage and airflow; avoid adding water (TC-20)
Low vegetative performance + low EC	Possible nutrient-related concern	Verify calibration and pH; seek expert advice before dosing

When observations conflict, for example an apparently low flower count with normal telemetry, the guidance requires visual inspection and a standardised recapture rather than selecting a cause (TC-22). The recommendation always contains the current stage, status, evidence, plausible issue, a low-risk step, urgency, next check and the limits of the evidence. Figure 4.7 shows the interaction for one checkpoint image.

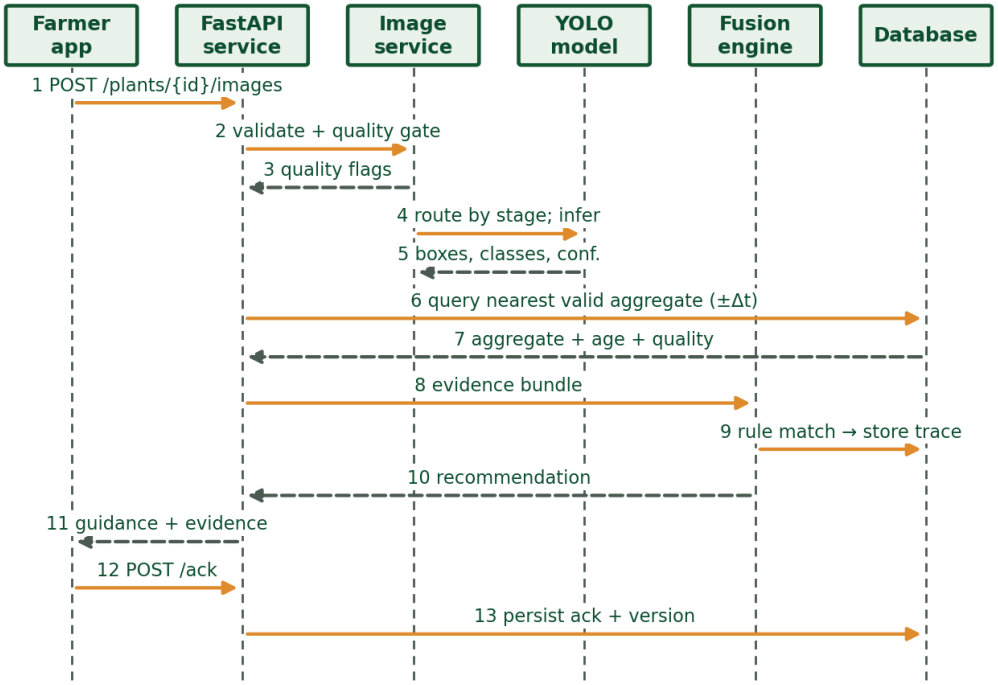


Figure 4.7. Sequence of events for a checkpoint image: validation, inference, alignment, fusion and acknowledgement.

4.9 Data Model and Interface Specification

Figure 4.8 shows the logical entities and Table 4.10 their purpose. Threshold profiles, rules and model versions are first-class entities so that every alert and recommendation points to the exact versions that produced it.

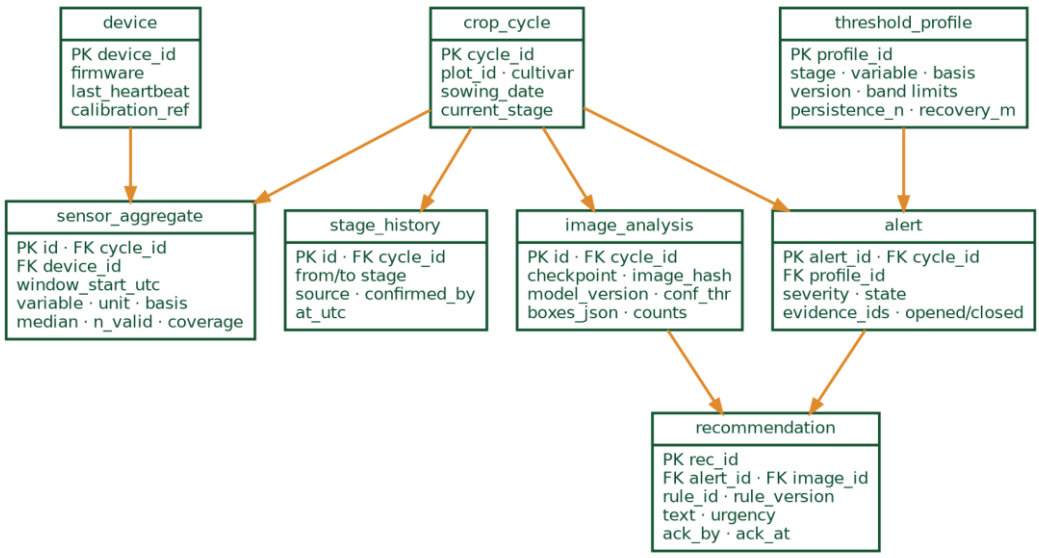


Figure 4.8. Logical data model (entities and key fields).

Table 4.10. Logical entities.

Entity	Key fields and purpose
crop_cycle	Plant, plot, cultivar, sowing date, current stage
stage_history	Previous and new stage, source (schedule, image, farmer), confirmer, time
sensor_aggregate	Message ID, device, UTC window, values, units, basis, quality, median, coverage
threshold_profile	Versioned stage targets, sensor basis, persistence and recovery parameters
image_analysis	Plant, checkpoint, image hash, model version, boxes, labels, visible counts, quality flags
alert / recommendation	Stage, severity, evidence IDs, rule and profile versions, advice, opened/closed, acknowledgement
device	Firmware, last heartbeat, connectivity, calibration reference

The HTTP interface follows a resource-oriented REST style [44]. The routes in Table 4.11 come from the implementation specification and must be reconciled with the repository; the alert-acknowledgement route is illustrative.

Table 4.11. Representative interface specification.

Method and path	Purpose	Auth / notes
POST /api/v1/iot/readings	Ingest validated telemetry aggregate	Device credential; idempotent by message_id
GET /api/v1/plants/{id}/live	Current values with freshness and quality	User token; owner only
GET /api/v1/plants/{id}/history	Time series for charts	User token; owner only; paged

Method and path	Purpose	Auth / notes
POST /api/v1/plants/{id}/images	Upload checkpoint image with metadata	User token; size and type limits
POST /api/v1/plants/{id}/analyze	Run stage-routed inference and fusion	User token; returns trace ID
GET /api/v1/plants/{id}/guidance	Current stage, status, evidence, advice	User token; owner only
POST /api/v1/plants/{id}/stage/confirm	Farmer confirms or corrects stage	User token; audited
POST /api/v1/alerts/{id}/acknowledge	Record acknowledgement (illustrative)	User token; stores actor, time, version

```
GET /api/v1/plants/P-001/guidance  ->  200 OK   (illustrative)
{ "stage": "flowering", "status": "watch", "evidence_quality": "good",
  "evidence": [ {"type":"sensor","ref":"agg-9841","note":"air_temp high for 20 min"},
                {"type":"image","ref":"img-5520","note":"14 visible flowers (this view)"} ],
  "interpretation": "Possible heat stress; not a diagnosis",
  "actions": ["Check shade/ventilation","Inspect flowers closely","Keep irrigation steady"],
  "next_check_min": 120, "rule": "R-FLW-HEAT-01@1.0", "profile": "FLW-v3" }
```

4.10 Mobile Interface Design

The journey comprises crop registration, live five-sensor cards, a five-stage timeline, due checkpoints, capture with a quality preview, detection results with an explanation, an action checklist, alert acknowledgement and history charts (Figure 4.9). Design follows Nielsen's heuristics [39]: visible system status (freshness and last valid time), match with the farmer's language (plain words, no model jargon), error prevention (capture guide and quality preview) and recognition rather than recall (icons with text). Status is never conveyed by colour alone. On invalid images the app requests recapture; on stale or missing sensors it shows the last valid timestamp and withholds unsafe recommendations; on an offline device it shows a connectivity notice distinct from crop-stress alerts; and when stage confidence is low it asks the farmer to confirm while retaining the machine suggestion. Sinhala and Tamil label sets are a readiness requirement (NFR-03).

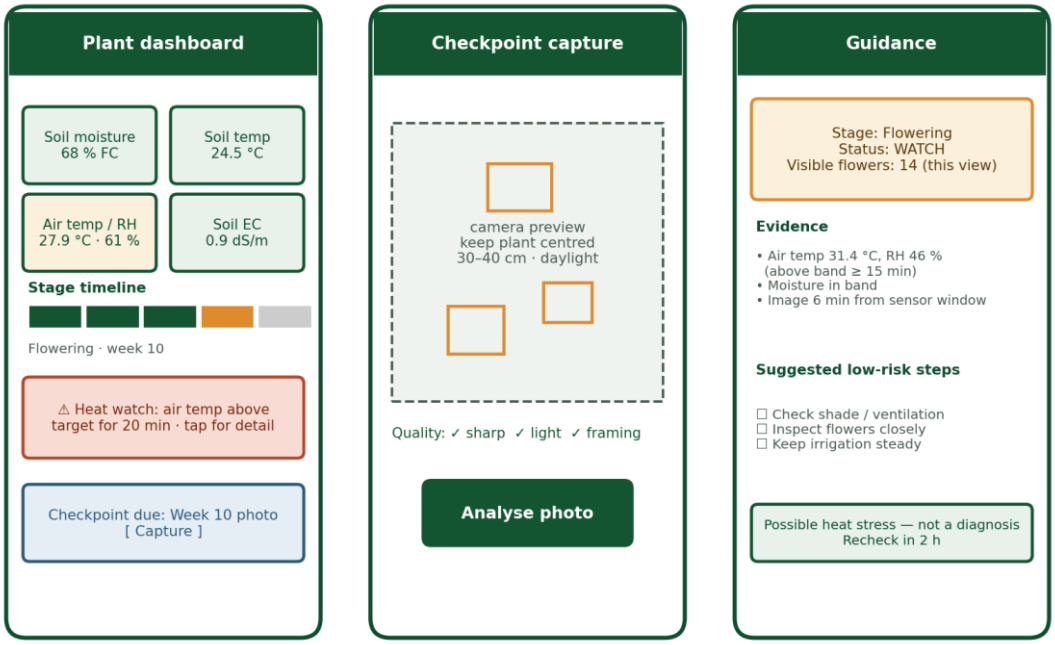


Figure 4.9. Low-fidelity wireframes of the dashboard, capture and guidance screens (illustrative values).

4.11 Security and Privacy Design

Table 4.12. Security threats and controls (guided by the OWASP API Security categories).

Threat	Control
Broken object-level authorisation (reading another farmer's plant)	Per-request ownership check on every plant, image and alert; denied attempts are logged (TC-25)
Broken authentication or leaked device credential	Token-based user authentication; unique per-device credentials that can be revoked; credentials kept out of firmware images and source control
Unrestricted resource consumption (large or repeated uploads)	File size and type limits; rate limiting; asynchronous inference queue
Excessive data exposure	Response schemas expose only required fields; coarse location only
Insecure transport or database rules	HTTPS everywhere; role-scoped, least-privilege database rules; separate staging project
Tampered model or rule files	Registry with checksums; changes versioned and reviewed

Privacy measures follow the data-minimisation and retention principles discussed in Section Section 3.10 [58], [59].

4.12 Verification and Validation Strategy

Test cases are specification-derived acceptance tests. They are not a claim that automated runs, field observations or acceptance results already exist; execution results, logs, commit identifiers and reviewer sign-off are entered in the evidence register (Appendix C) when tests are performed, and are reported in the results chapter. A test passes only when the observed output matches the asserted expected output; otherwise it is recorded as failed or blocked with a reason.

4.12.1 Test levels and environment

Testing is organised in four levels: unit tests (classification, persistence, alignment, ratio arithmetic), component tests (ingestion, quality gate, model routing), integration tests (ESP32 or simulator to Firebase to API to app) and scenario tests (end-to-end checkpoint runs). The environment uses an authenticated simulator or controlled ESP32 node, a staging Firebase project, a versioned FastAPI service, seeded crop-cycle records, separately labelled detector checkpoints, fixed UTC timestamps with an explicit Asia/Colombo display conversion, deterministic fixtures and pinned profile versions. No test may trigger a real plant-care action. For model tests the image hash, ground-truth annotation, checkpoint hash, confidence setting, library version and visible counts are stored.

4.12.2 Core functional and failure-mode test cases

Table 4.13. Acceptance test cases (specification-derived; execution status recorded in Appendix C).

ID	Level	Scenario	Expected result (acceptance criterion)	Req.
TC-01	Component	Valid signed payload with five values and UTC time	Accepted; stored with correct units, device and plant IDs	FR-02
TC-02	Component	Duplicate message_id sent twice	One logical reading; duplicate acknowledged idempotently	FR-02
TC-03	Unit	Soil moisture below zero or implausibly high	Flagged invalid; no stress diagnosis from it	FR-10
TC-04	Unit	EC supplied without measurement basis	Not compared with ECe limits; basis requested	FR-10
TC-05	Component	Last valid reading older than freshness window	Stale flag; confidence reduced; missing is not zero	FR-10
TC-06	Unit	Confirmed germination plant requests check	Germination profile used; no image mandatory	FR-01
TC-07	Unit	Confirmed seedling plant requests check	Seedling profile; no fruit detector called	FR-01, FR-05
TC-08	Component	Vegetative plant at Week 7 checkpoint	Capture guidance and correct route or availability notice	FR-05

ID	Level	Scenario	Expected result (acceptance criterion)	Req.
TC-09	Component	Flowering plant at Week 10 or 11	Flower workflow; count labelled view-limited	FR-05
TC-10	Component	Fruiting plant at Week 16 or 20	YOLO11s classes used; total and ripening ratio consistent	FR-05
TC-11	Unit	Value exactly on each target limit	Deterministic band per inclusive rule	FR-07
TC-12	Unit	One isolated out-of-range reading	No persistent alert before n samples	FR-07
TC-13	Unit	Repeated abnormal samples after alert opens	Same alert updated; no duplicates during cooldown	FR-07
TC-14	Unit	m valid in-band samples after alert	Alert resolves once; recovery logged	FR-07
TC-15	Integration	ESP32 stops reporting beyond threshold	Connectivity notice, not crop-stress diagnosis	FR-11
TC-16	Component	Blurred or dark flower photograph	Quality warning and recapture; no reliable count	FR-04
TC-17	Component	Flower image sent while crop is fruiting	Stage mismatch shown; unrelated detector not used silently	FR-05
TC-18	Unit	Image and aggregate separated beyond tolerance	Sensor marked non-concurrent; fused conclusion withheld	FR-06
TC-19	Unit	Low moisture + high air temperature + low RH	Possible drought/heat; cautious inspection advice	FR-07
TC-20	Unit	High moisture + high RH	Drainage and ventilation review; no advice to add water	FR-07
TC-21	Unit	Low moisture + elevated EC	Possible concentration effect; verify basis and water quality	FR-07, FR-10
TC-22	Scenario	Week 10 and 11 images with mismatched view	No flower-drop rate asserted; recapture requested	FR-04
TC-23	Unit	12 visible fruits, 3 classified ripening	Ratio 25 % of visible fruits; not farm yield	FR-05
TC-24	Component	Low-confidence or heavily occluded fruit detections	Uncertainty shown; absence not assumed	FR-04
TC-25	Integration	Unauthenticated or other-farmer read attempt	Access denied; event logged; no data leaked	NFR-05
TC-26	Integration	Firebase unreachable then device reconnects	Retry and buffer; no duplicate records	FR-11
TC-27	Integration	Farmer acknowledges a recommendation	Timestamp, actor and version retained	FR-09
TC-28	Integration	Authorised manual stage correction	History keeps previous/current and source; next profile selected	FR-12
TC-29	Scenario	Dashboard with units and last-updated time	Correct unit, stage range and freshness shown	FR-03, FR-08
TC-30	Scenario	Valid fruit photo + concurrent five-sensor aggregate	Detection summary, bands, reasoning and safe actions persisted and shown	FR-08, FR-09

4.12.3 Model evaluation and acceptance protocol

Images are split by plant, visit or session before augmentation so that near-identical crops never appear in both training and test. For each detector the protocol reports

precision, recall, F1, mAP@0.50 and mAP@0.50:0.95 on the untouched test set with the IoU and confidence thresholds stated (Equations 3.1 to 3.3), and count agreement by MAE, RMSE and bias (Equation 3.4). For ripeness, a class confusion matrix, per-class F1 and the error in the visible ripening ratio are given, counting only quality-approved comparable photographs. Results are reported separately for the seedling, flowering and fruiting datasets, and the approximately 118-image dataset is not described as 118 independent farm observations without reviewing provenance and augmented duplicates.

Table 4.14. Evaluation items, reference evidence, metrics and reporting safeguards.

Item	Reference evidence	Metrics	Safeguard
YOLOv8n seedling emergence	Manually labelled independent seedling images	mAP50–95, recall, count MAE	Confirm checkpoint and class names
YOLO flowering detection	Labelled flower and young-fruit images	Per-class AP, count MAE, false detections	Compare identical plant framing
YOLO11s fruit/ripening	Independent annotated ripening and unripe images	mAP50–95, class F1, count MAE, ratio error	Report occlusion and viewpoint limits
Sensor subsystem	Calibrated reference instruments	MAE, bias, limits of agreement, stale rate	Do not equate raw ADC with %FC
Recommendation logic	Agronomist-reviewed blinded cases	Agreement, unsafe-recommendation rate	Avoid causal diagnosis from correlation alone
Full workflow	Timestamped staging runs and app captures	Latency, completion, API failures	Separate lab and simulation from field results

4.12.4 Requirement-to-test traceability

Table 4.15. Traceability from design requirement to verification evidence.

Design requirement	Test IDs	Evidence needed
ESP32 to Firebase telemetry	TC-01 to TC-05, TC-15, TC-26	Raw payloads, database snapshots, calibration metadata
Five-stage crop monitoring	TC-06 to TC-11, TC-28	Stage history and profile-version tests
Persistent alert policy and lifecycle	TC-12 to TC-15	Timestamped alert history and notification records
YOLOv8/YOLO11 inference and image scheduling	TC-08 to TC-10, TC-16 to TC-18, TC-22 to TC-24	Checkpoint hashes, predictions, labelled images
Explainable image–IoT fusion	TC-18 to TC-21, TC-30	Decision trace, concurrent sensor window, output
Privacy and farmer-facing clarity	TC-25, TC-27, TC-29	Permission checks, UI captures, reviewer notes

Threats to the validity of these tests, and their mitigations, are consolidated in Table 3.6. In particular, a bench-calibrated EC probe in a nutrient solution must not be assumed equivalent to saturated-paste ECe, and the stage ranges remain provisional design profiles rather than universal *C. chinense* limits.

4.13 Individual Contribution and Group-Interface Boundaries

Savinda R.M.D (IT22124630) is responsible for the A-to-Z growth monitoring and guidance subsystem: the ESP32/Firebase monitoring workflow, stage-dependent interpretation, stage-specific YOLOv8 and YOLO11 visual evidence, and explainable farming alerts and actions. The Pre-Analysis and Smart Decision Support component provides upstream farm, plot and cultivation-plan context. Outputs of this component are associated with a registered plant and crop cycle and may be consumed by wider NAIMIRIS interfaces. The independent pest and disease detection component belongs to another researcher and is neither claimed nor assessed here.

4.14 Chapter Summary

The design converts the methodology into a concrete, traceable system: validity-first sensing, stage-aware profiles with persistence-based alerts, stage-routed detectors with a quality gate, rule-based fusion with decision traces, a versioned data model, a farmer-oriented interface and a thirty-case verification suite linked to requirements. The remaining work, reported in later chapters, is to execute the protocol: collect independent local data, calibrate the sensors, train and test the detectors on sealed partitions, and run the scenario, usability and expert evaluations.

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GLOSSARY

Stage-aware monitoring. Evaluating a measurement against a target profile defined for the current crop-growth stage.

Visible-object count. The number of organs recognisable in a photograph; not necessarily the total on the plant.

Data fusion. Joint interpretation of visual detections, sensor readings, temporal context and data-quality indicators.

Sensor calibration. Establishing a defensible relationship between raw measurements and meaningful physical units.

Persistence (on-delay). A requirement that a deviation continue for a set number of valid samples before an alert opens.

Recovery (off-delay). A requirement that in-band samples continue for a set number before an alert resolves.

Plant-disjoint split. A partition of images in which all images of a given plant or session fall in exactly one of the training, validation or test sets.

Decision trace. The stored record of rule, profile and model versions, and evidence identifiers behind a recommendation.

APPENDIX A: CLAIM TRACEABILITY

This appendix records the source artefact for each class of statement and how it is treated in the thesis.

Table A.1. Claim-to-source distinctions.

Content	Source artefact	Treatment
Five-stage workflow and ESP32/Firebase monitoring	PP2, pp. 15–16	Reported functionality
YOLOv8n seedling, flowering YOLO and YOLO11s fruit models	PP2, p. 17	Reported model types; benchmarks not supplied
Checkpoint weeks; schema and thresholds	Implementation report, pp. 2–12	Design specification; local validation required
Original four-week YOLOv8/ResNet-18 design	March proposal, pp. 3 and 18–21	Historical proposed architecture
Exclusion of pest and disease detection	PP2, pp. 13–19 (other student component)	Out of scope
Literature added after the proposal	Peer-reviewed and standards sources (reference list)	Not new measured experiments

APPENDIX B: ACADEMIC INTEGRITY, AI-USE DISCLOSURE AND EVIDENCE CHECKLIST

AI-use disclosure. Generative AI assistance was used in preparing this document for restructuring, drafting, language editing, diagram production and reference formatting. The author has the responsibility to verify all technical statements, citations and project details against the original records and to confirm this disclosure against the SLIIT policy before submission. No accuracy figure, field-trial result, supervisor approval or test execution is asserted anywhere in the document.

Evidence to attach before the viva: raw detector test reports for YOLOv8n, the flowering detector and YOLO11s; training settings, validation split and label maps; dataset manifest and licences; sensor calibration records; end-to-end test logs; dashboard and API captures; alert-logic test results; image-count agreement samples; security configuration; and any consent approvals and stakeholder feedback. Replace provisional requirements wherever the implementation differs.

APPENDIX C: TEST EXECUTION REGISTER (TEMPLATE)

This register is a template. Entries are completed when tests are executed; no test is recorded as passed until its observed output matches the expected result.

Table C.1. Specimen execution log.

Run ID	Linked case	Date	Observed output / evidence link	Result	Tester sign-off
RUN-01	TC-01	Not recorded	To be attached	NOT EXECUTED	Pending
RUN-02	TC-12	Not recorded	To be attached	NOT EXECUTED	Pending
RUN-03	TC-18	Not recorded	To be attached	NOT EXECUTED	Pending
RUN-04	TC-23	Not recorded	To be attached	NOT EXECUTED	Pending
RUN-05	TC-30	Not recorded	To be attached	NOT EXECUTED	Pending
RUN-06	TC-25	Not recorded	To be attached	NOT EXECUTED	Pending